



**Faculty of Management and Business
Intelligence
Business Informatics Department**



**AI Driven Health and Lifestyle Monitoring With Python and Power
BI**

A Graduation Research Project Submitted in Partial Fulfilment of The
Requirements for The Degree of Bachelor of Management Information
Systems

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Ahmed Hatem, Ahmed Taher, Mariem Hassan, Nour Eldeen, Yousef Ibrahim

DEDICATION

To our beloved families,
whose unconditional love and support have been our greatest strength.

To our teachers and mentors,
who inspired us to pursue knowledge and excellence.

To our friends and colleagues,
for the shared moments and encouragement along this journey.

DECLARATION

We hereby declare that this graduation project AI Driven Health and Lifestyle Monitoring with Python and Power BI is our own original work. It has been submitted in partial fulfilment of the requirements for the degree of Bachelor of Science in Management Information Systems at the Faculty of Management and Business Intelligence.

We confirm that:

- This project has not been submitted for any other degree or award at this or any other institution.
- All sources of information have been properly cited and referenced.
- Intellectual property of others has been duly acknowledged.

ABSTRACT

With the rapid advancement of information technology and digital transformation, organizations increasingly rely on intelligent information systems to improve their operational efficiency and decision-making capabilities.

AI-driven health and lifestyle monitoring has become essential as individuals increasingly seek proactive ways to manage well-being and prevent chronic conditions. This project addresses the challenge of fragmented and underutilized health data by proposing an integrated system that leverages Python for data processing and machine learning, alongside Power BI for interactive visualization and reporting. The system collects and consolidates lifestyle and health-related data—such as sleep patterns, physical activity, and basic health indicators—then preprocesses it to ensure quality and consistency. Using Python-based analytical techniques, including data cleaning, feature engineering, and predictive modeling, the system identifies patterns and generates health risk predictions. These insights are then visualized through dynamic Power BI dashboards, enabling users and healthcare managers to monitor key metrics, explore trends, and make informed decisions. The methodology combines exploratory data analysis, supervised learning models, and performance evaluation to ensure reliability and accuracy. Major findings indicate that integrating machine learning with visual analytics significantly improves the ability to detect early signs of health risks and enhances user engagement with personal health data. The project concludes that a unified, data-driven approach to health monitoring can support better lifestyle choices, enable early intervention, and provide a scalable solution for both individual users and healthcare

professionals, ultimately contributing to improved overall well-being and preventive healthcare practices.

Keywords: AI-driven healthcare, lifestyle monitoring, machine learning, data visualization, predictive analytics

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CHAPTER ONE: GENERAL FRAMEWORK

1.1 Introduction

The development of information systems has become one of the most critical factors in the success of modern organizations. Business Information Systems (BIS) integrate technology, people, and processes to provide organizations with the information they need to make effective decisions. The convergence of artificial intelligence (AI) with ubiquitous sensing technologies has ushered in a transformative era for health and lifestyle monitoring. Wearable devices ranging from smartwatches and fitness trackers to smart clothing and medical-grade patches have become increasingly prevalent, capable of continuously capturing diverse physiological signals such as heart rate, electrocardiogram (ECG), photoplethysmography (PPG), blood pressure, oxygen saturation, body temperature, and motion data. These devices, often integrated with smartphone-based mHealth platforms and cloud-edge computing infrastructure, generate high-frequency, multimodal data streams that hold immense potential for real-time health assessment, early disease detection, personalized interventions, and proactive lifestyle management. The COVID-19 pandemic acted as a significant catalyst, accelerating the adoption of AI-powered mHealth applications for remote monitoring, early detection, and telehealth services. This experience expanded the role of AI beyond traditional clinical settings, demonstrating its capacity to support population-scale health surveillance and decentralized care delivery. Simultaneously, rapid advances in machine learning (ML) have dramatically enhanced the analytical capabilities of health monitoring systems, enabling automated feature extraction, pattern recognition, predictive modelling, and anomaly detection from complex, high-dimensional physiological

signals. Despite these technological breakthroughs, the widespread adoption and clinical integration of AI-driven health monitoring systems remain constrained by several persistent challenges. Technical issues such as device heterogeneity, sensor noise, signal artifacts, and distribution shifts across populations and environments undermine model robustness and generalizability. Ethical and regulatory concerns surrounding data privacy, algorithmic bias, informed consent, model opacity, and legal liability raise critical barriers to clinical deployment. Furthermore, most existing systems have been validated in controlled laboratory settings rather than real-world environments, with limited large-scale, longitudinal clinical trials demonstrating tangible improvements in patient outcomes. The translation from proof-of-concept to clinical practice thus remains a significant hurdle. This literature review aims to provide a comprehensive synthesis of current research on AI-driven health and lifestyle monitoring systems. It seeks to: (i) map the technological landscape, including sensor modalities, data acquisition, and computational infrastructure; (ii) categorize and evaluate AI/ML methodologies employed in health monitoring applications; (iii) examine clinical and lifestyle applications across key disease domains and behavioural interventions; (iv) identify and critically appraise the major challenges and limitations reported in the literature; and (v) propose future research directions to accelerate the safe, equitable, and effective deployment of AI-driven monitoring systems in healthcare.

1.2 Problem Analysis

Modern life has created a major problem called “Self-Neglect” among students and professionals. Many people focus heavily on work, studying, and productivity while ignoring important parts of their health, such as sleep quality and physical activity. Over time, this can lead to stress, poor performance, unhealthy routines, and serious long-term health problems. These unhealthy habits may gradually increase the risk of chronic diseases, especially heart diseases and sleep disorders. One of the biggest issues is the lack of early awareness, as most health monitoring tools only record problems after they occur instead of predicting risks before they become serious. As a result, many individuals do not realize that their daily lifestyle may negatively affect their future health until a medical problem appears. This project addresses this issue by developing a Machine Learning prediction model using techniques such as Random Forest and Logistic Regression to identify hidden health patterns from daily habits. By providing early disease predictions before symptoms become severe, the model helps individuals make healthier decisions, reduce future health risks, and improve their overall quality of life.

1.3 Objectives of the Project

The primary objectives of this project are as follows:

- To analyse the existing Data Set and identify its shortcomings and areas for improvement.
- To design and develop a comprehensive Predictive Model that addresses identified problems.
- To implement and test the proposed Predictive Model using appropriate technologies.
- Develop Predictive Model to integrate health and lifestyle data from multiple sources (e.g., sleep, activity, and basic health metrics).
- Apply data preprocessing techniques to clean, normalize, and handle missing or inconsistent data.
- Build and compare multiple machine learning models to improve prediction accuracy for health risks.
- Enhance model interpretability to provide clear and understandable insights for users and healthcare professionals.
- Design interactive dashboards using Power BI for data visualization and monitoring.
- Enable users to track key health indicators and identify patterns or trends.
- Generate health reports for better decision-making.

- Improve user engagement by delivering personalized insights and recommendations.
- Support proactive healthcare by enabling early detection of potential health issues.

1.4 Significance of the Project

This project carries significant academic, practical, and organizational value. From an academic perspective, it contributes to the body of knowledge in business information systems by applying theoretical concepts to a real-world problem. This project matters because it transforms how health and lifestyle data are used from passive tracking to proactive, insight-driven decision-making creating value for organizations, end users, and the broader community. For organizations, especially healthcare providers and wellness programs, the system enables more efficient and informed decision-making. By integrating data analytics and machine learning with visualization tools like Power BI, organizations can monitor population health trends, identify high-risk groups earlier, and allocate resources more effectively. This reduces operational costs, improves service quality, and supports a shift from reactive treatment to preventive care. For end users, the project offers a practical and personalized way to understand and improve their health. Instead of viewing isolated metrics, users gain access to meaningful insights, predictions, and clear visual dashboards that highlight patterns in sleep, activity, and overall lifestyle. This empowers individuals to make better daily decisions, adopt healthier habits, and detect potential health risks before they become serious problems. On a broader scale, the project contributes to public health by promoting awareness and prevention. When individuals are better informed and more engaged

with their health data, it can lead to a reduction in lifestyle-related diseases, easing the burden on healthcare systems and improving overall community well-being. What makes this project innovative is its integration of multiple components into a single, cohesive system. It combines data preprocessing, predictive modeling using Python, and real-time interactive visualization, rather than treating them as separate processes. Additionally, the focus on interpretability and user-friendly design ensures that advanced analytics are accessible, not just to data experts, but to everyday users. This blend of AI-driven prediction with intuitive dashboards creates a scalable and forward-thinking solution for modern healthcare challenges.

1.5 Methods

1. Data Collection

Data collection is the initial phase of the data analysis workflow in which relevant data is gathered from various sources to support the objectives of the study or system. The collected data may originate from databases, wearable devices, sensors, mobile applications, surveys, healthcare records, or online platforms. This stage is considered critical because the quality, accuracy, and completeness of the collected data directly influence the reliability of the analysis results. During this phase, organizations ensure that the data sources are trustworthy, consistent, and capable of providing sufficient information for analysis. Proper data collection also involves defining the type of data required, the frequency of collection, and the methods used to store and manage the data securely.

2. Data Cleaning and Pre-processing

Data cleaning and preprocessing involve preparing raw data for analysis by identifying and correcting errors, inconsistencies, and incomplete records. Real-world data is often noisy, duplicated, or missing important values, which may negatively affect the accuracy of analytical models. Therefore, preprocessing techniques such as handling missing values, removing duplicate records, normalization, encoding categorical variables, and data transformation are applied to improve data quality. This stage ensures that the dataset becomes structured, organized, and suitable for analytical and machine learning processes. Effective preprocessing increases the efficiency of the system and enhances the reliability of the final outcomes.

3. Exploratory Data Analysis (EDA)

Exploratory Data Analysis is the process of examining and understanding the characteristics of the dataset before applying advanced analytical models. In this stage, statistical methods and visualization techniques are used to identify patterns, trends, correlations, and anomalies within the data. Common tools include graphs, histograms, scatter plots, and correlation matrices. EDA helps researchers and analysts gain deeper insights into the dataset and supports informed decision-making regarding feature selection and model design. Furthermore, this stage assists in detecting unusual

behavior or hidden relationships that may significantly influence the analytical results.

4. Feature Selection and Engineering

Feature selection and engineering focus on identifying the most relevant variables that contribute effectively to the predictive performance of the analytical model. In many datasets, not all variables are useful, and some may introduce noise or reduce model accuracy. Feature selection techniques are therefore applied to retain only significant attributes, while feature engineering involves creating new variables from existing data to improve analytical capabilities. This phase enhances model efficiency, reduces computational complexity, and improves prediction accuracy. Proper feature engineering is considered an essential step in machine learning and predictive analytics applications.

5. Data Modelling and Analysis

Data modeling and analysis represent the core phase of the workflow, where statistical and machine learning techniques are applied to extract meaningful insights and predictions from the data. Different analytical methods such as regression, classification, clustering, and time-series forecasting may be used depending on the project objectives. Machine learning algorithms are trained using historical data to identify patterns and predict future outcomes. In healthcare and lifestyle monitoring systems, this stage may be used to detect health risks, classify activity

patterns, or predict abnormal conditions. The effectiveness of this stage depends on the quality of the dataset and the suitability of the chosen analytical methods.

6. Model Evaluation and Validation

Model evaluation and validation are conducted to measure the performance, reliability, and accuracy of the developed analytical models. Various evaluation metrics such as accuracy, precision, recall, F1-score, and mean squared error are used depending on the type of analysis performed. Validation techniques such as cross-validation and train-test splitting help ensure that the model generalizes effectively to new and unseen data. This phase is essential for identifying weaknesses, reducing over fitting, and improving the overall quality of the system. A well-evaluated model increases confidence in the predictive outcomes and supports reliable decision-making.

7. Data Visualization and Reporting

Data visualization and reporting involve presenting analytical results in a clear and understandable form to support interpretation and decision-making. Visualization tools such as dashboards, charts, graphs, and heatmaps are commonly used to display trends, comparisons, and predictions interactively. Reporting transforms technical analytical

findings into meaningful information that can be understood by stakeholders, healthcare professionals, and decision-makers. This stage enhances communication between technical and non-technical users and enables organizations to make informed strategic decisions based on data-driven insights.

Data Collection Techniques:

Data was collected through Kaggle datasets and simulated health records, including sleep patterns, physical activity levels, and general lifestyle indicators. This data was then cleaned and preprocessed to ensure consistency before being used in analysis and modeling.

Tools and Technologies Used

- Python: Core language for data processing and machine learning
- Pandas: Data cleaning and manipulation
- Scikit-learn: Machine learning model development
- Power BI: Interactive dashboards and visualization
- Jupiter Notebook: Development and experimentation environment

1.6 Delimitation and Scope of Study

The scope of this project is focused on developing an AI-driven health and lifestyle monitoring system using Python and Power BI, aimed at analyzing and predicting health-related patterns based on lifestyle data such as sleep, activity levels, and general well-being indicators. The study is limited to system design, implementation, and evaluation using simulated or publicly available datasets rather than real-time hospital or clinical data.

Organizational Scope:

This project is developed for academic and research purposes within a healthcare analytics context. It is intended for use by individuals such as healthcare managers and data analysts who require insights into lifestyle and health trends for decision-making and monitoring purposes.

Technical Scope:

The system is implemented using Python for data preprocessing, analysis, and machine learning model development. Predictive analytics is performed using libraries such as Pandas, and Scikit-learn. Data visualization and dashboard creation are handled using Power BI. The system focuses on batch data processing and does not include real-time IoT integration or mobile application development.

Time Scope:

The project is carried out within the academic year 2025–2026, covering all phases from requirements analysis and system design to implementation, testing, and documentation.

Delimitations:

- The system does not include real-time data streaming from wearable devices.
- It is limited to selected health and lifestyle parameters such as sleep, activity, and basic health indicators.
- The predictive models are trained on available datasets and may not represent clinical-grade medical diagnosis.
- The system is designed as a prototype and does not include full-scale deployment in healthcare institutions.

1.7 Terms and Definitions

The following key terms are used throughout :

Term	Definition	Context in Project
Artificial Intelligence (AI)	The simulation of human intelligence processes by machines, especially computer systems.	Used to analyze health and lifestyle data and generate predictive insights.
Machine Learning (ML)	A branch of AI that uses algorithms to learn from data and make predictions without explicit programming.	Used to build predictive models and risk detection.
Data Proprocessing, Predictive Analytics	The process of cleaning, transforming, and organizing raw data into a usable format for analysis.	Applied to improve diagnostic and risk modeling.
Data Visualization, Dashboard	The graphical representation of information and data using charts, graphs, and dashboards.	Implemented as the primary tool for interactive data presentation.
Dataset	A structured collection of labeled datasets used for analysis and model training.	Utilized to gather health and lifestyle metrics.
Pattern Recognition	The process of identifying and understanding variable patterns from health-related measurements to provide personalized performance insights.	Used in conjunction with machine learning algorithms to detect lifestyles and health risks.
Health Monitoring	The process of collecting and tracking health-related data to assess well-being.	Utilized to gather health and lifestyle metrics.
Health Monitoring	The process of collecting and tracking health data.	Core purpose of the project system.

Table 1.1: Key Terms and Definitions

CHAPTER TWO: LITERATURE REVIEW

2.1 Literature Review

Introduction: A New Era in Health Tracking

We are living through a major shift in how we think about health (Morelli et al., Not long ago, checking your health meant going to a doctor's office for a one-time snapshot your blood pressure, heart rate, or blood sugar at that exact moment (Braunwald, 2015). But today, thanks to smartwatches, fitness trackers, and phone apps, we can monitor our bodies continuously, 24/7 (Patel, 2026). These wearable devices track everything from heart rate and sleep patterns to stress levels and physical activity (Smart Healthcare, 2025). This constant stream of data, combined with artificial intelligence (AI), opens up exciting possibilities (Aversano et al., 2024). Instead of just seeing what happened yesterday, AI can help predict what might happen tomorrow like spotting early signs of a sleep disorder or heart problem before symptoms appear (Zhang et al., 2024). With hospitals overwhelmed and people stuck at home, remote monitoring became a necessity, not a luxury (Morelli et al., 2025). This review looks at how AI is changing health and lifestyle monitoring (How are Machine Learning and Artificial Intelligence Used in Digital Behavior Change Interventions, 2024). We will explore the technology behind it, the methods used, real-world applications, and the tools like Python and Power BI that make it all work (Davenport & Harris, 2017). The goal is to understand where we are, how we got here, and where we are going (Exploring the Applications of Explainability in Wearable Data Analytics, 2025).

2.2 How We Got Here: A Short History of Health Monitoring

To understand the present, it helps to look at the past (IOM, 1991). Health monitoring has gone through four main phases (Shortliffe & Cimino, 2014). **Phase one:** the clinic era (Braunwald, 2015). In the 19th and early 20th centuries, tools like the stethoscope (1816), blood pressure cuff (1896), and EKG machine (1903) revolutionized medicine (Braunwald, 2015). For the first time, doctors could measure what was happening inside the body (NASA, 1965). But these measurements were only taken during clinic visits brief snapshots that missed what happened the rest of the time (IOM, 1991). **Phase two:** going digital (Shortliffe & Cimino, 2014). The 1980s and 1990s brought electronic health records (EHRs) (IOM, 1991). Patient information moved from paper files to computers (Shortliffe & Cimino, 2014). This made storage and retrieval easier, but these systems were built for billing and administration, not for continuous monitoring or prediction (IOM, 1991). Data also stayed locked inside each hospital or clinic, unable to travel with the patient (Shortliffe & Cimino, 2014). **Phase three:** wearables for everyone (Patel, 2026). The real game-changer came in the 2000s with consumer devices like Fitbit (2009) and the Apple Watch (2015) (Patel, 2026). Suddenly, millions of people could track their steps, heart rate, and sleep from their wrists (Shcherbina et al., 2017). This generated massive amounts of real-world health data (Patel, 2026). However, early wearables had problems: they weren't always accurate, lacked clinical validation, and gave users lots of numbers but little guidance on what to do with them (Shcherbina et al., 2017). **Phase four:** AI enters the picture (Aversano et al., 2024). The 2010s and 2020s have been about adding intelligence (Aversano et al., 2024). Machine learning algorithms can now find patterns in all that wearable data predicting heart problems, detecting sleep apnea, or forecasting blood sugar swings (Zhang et al., 2024). This

is where we stand today: not just collecting data, but actually making sense of it (Patel, 2026).

2.3 Three Generations of Data Analytics in Health

Alongside the hardware evolution, the way we analyze health data has gone through three generations (Sanders, 2014). First generation: descriptive (what happened?) (Sanders, 2014). Early fitness apps simply showed you your step count or average heart rate (Sanders, 2014). You could see you slept poorly last night, but not why (Åkerstedt et al., 2014). Helpful, but limited (Sanders, 2014). Second generation: diagnostic (why did it happen?) (Åkerstedt et al., 2014). Next came tools that could find relationships (Åkerstedt et al., 2014). For example, researchers found a strong link between high stress and poor sleep (correlation of -0.89, meaning more stress strongly predicts worse sleep) (Åkerstedt et al., 2014). Dashboards like Tableau and Power BI let users explore these connections visually (Davenport & Harris, 2017). Still, these tools mostly looked backward, not forward (Few, 2012). Third generation: predictive (what will happen? and what should I do?) (Patel, 2026). This is the current frontier (Patel, 2026). Machine learning models can predict your risk of a sleep disorder based on your age, stress, activity, and BMI (Zhang et al., 2024). One study achieved 95.5% accuracy predicting sleep apnea and insomnia using a random forest model (Zhang et al., 2024). And when you combine these predictions with an easy to use dashboard, anyone not just data scientists can get actionable insights (Davenport & Harris, 2017).

2.4 What Other Researchers Have Done (And What's Still Missing)

Let us look at ten key studies that shaped this field (Esteva et al., 2017; Rajkomar et al., 2018; Wang et al., 2019; How are Machine Learning and Artificial Intelligence Used in Digital Behavior Change Interventions? 2024; Exploring the Applications of Explainability in Wearable Data Analytics, 2025; Federated Learning in Smart Healthcare, 2025; Morelli et al., 2025; Patel, 2026; Wearable Artificial Intelligence for Sleep Disorders, 2025; Radanliev, 2025).

Study 1: skin cancer detection with deep learning (Esteva et al., 2017). Researchers trained an AI to spot skin cancer from photos (Esteva et al., 2017). It performed as well as dermatologists (Esteva et al., 2017). But it needed high-quality clinical images and did not use continuous wearable data (Esteva et al., 2017). Our project focuses on everyday lifestyle data from wearables, not specialized medical images (Patel, 2026).

Study 2: predicting hospital outcomes from EHRs (Rajkomar et al., 2018). This study used deep learning on millions of electronic health records to predict who would die or be readmitted (Rajkomar et al., 2018). It worked very well, but required massive clinical datasets that individuals or small clinics do not have (Rajkomar et al., 2018). Our project uses simple lifestyle data anyone can collect (Patel, 2026).

Study 3: wearable-based real-time alerts (Wang et al., 2019). Researchers built a system that used smartwatch data to detect irregular heart rhythms (Wang et al., 2019). It worked, but only at 82% accuracy compared to medical-grade EKGs (Wang et al., 2019). It also lacked predictive analytics and good dashboards (Wang et al., 2019). Our project adds machine learning predictions and Power BI visualizations (Davenport & Harris, 2017).

Study 4: AI in digital behavior change (How are Machine Learning and Artificial Intelligence Used in Digital Behavior Change Interventions?, 2024). This review of 89 studies found that AI can moderately improve physical activity and sleep hygiene (effect sizes of 0.4-0.6) (How are Machine Learning and Artificial Intelligence Used in Digital Behavior Change Interventions?, 2024). Random forests were among the most successful methods (How are Machine Learning and Artificial Intelligence Used in Digital Behavior Change Interventions?, 2024). However, most studies were short-term (under 12 weeks) and had small sample sizes (How are Machine Learning and Artificial Intelligence Used in Digital Behavior Change Interventions?, 2024). Study 5: explainable AI for wearables (Exploring the Applications of Explainability in Wearable Data Analytics, 2025).

Researchers found that making AI models explainable (using tools like SHAP and LIME) improved user trust by 35-50% (Exploring the Applications of Explainability in Wearable Data Analytics, 2025). Showing which features mattered most for example, that stress level was the strongest predictor of sleep quality was especially effective (Exploring the Applications of Explainability in Wearable Data Analytics, 2025). Our project includes clear feature importance metrics and intuitive Power BI visuals (Exploring the Applications of Explainability in Wearable Data Analytics, 2025).

Study 6: federated learning for privacy (Federated Learning in Smart Healthcare, 2025). Federated learning allows AI models to train on data from multiple hospitals without ever sharing the raw patient data (Federated Learning in Smart Healthcare, 2025). This preserves privacy while still learning from large, diverse datasets (Federated Learning in Smart Healthcare, 2025). Our current project uses a centralized approach, but federated learning is a promising future direction (Federated Learning in Smart Healthcare, 2025).

Study 7: AI- driven mobile health (Morelli et al., 2025).

This review showed the strongest evidence for AI in cardiovascular monitoring, diabetes management, and mental health (Morelli et al., 2025).

However, many studies lacked real- world clinical validation (Morelli et al., 2025).

Our project contributes by validating a sleep disorder prediction system on realistic lifestyle data (Morelli et al., 2025).

Study 8: AI- integrated wearables (Patel, 2026).

Patel found that wearable- based AI could detect atrial fibrillation with 90- 95% sensitivity and sleep disorders with 80- 85% accuracy (Patel, 2026).

The pandemic greatly accelerated adoption of such systems (Patel, 2026).

Our project achieves 95.58% accuracy for sleep disorder prediction, responding to Patel's call for real- world validation (Patel, 2026).

Study 9: wearable AI for sleep disorders (Wearable Artificial Intelligence for Sleep Disorders, 2025).

This review identified random forests and gradient boosting as the top performers for sleep disorder classification, with 85- 92% accuracy (Wearable Artificial Intelligence for Sleep Disorders, 2025). Key predictors included heart rate variability, stress, and lifestyle factors (Wearable Artificial Intelligence for Sleep Disorders, 2025). Our project uses a random forest model that achieves 95.58% accuracy, and we provide actionable dashboards for healthcare managers (Wearable Artificial Intelligence for Sleep Disorders, 2025).

Study 10: privacy and ethics (Radanliev, 2025).

Radanliev found that only 30% of wearable AI systems had clear privacy policies, and 25% showed algorithmic bias (e.g., lower accuracy for darker skin tones) (Radanliev, 2025). Model opacity remains a major barrier to clinical deployment (Radanliev, 2025). Our project addresses this by using an interpretable random forest model and providing clear feature importance, along with documented privacy considerations (Radanliev, 2025).

2.5 Theoretical Background

Our project is grounded in several well-established theories (Pressman, 2015; Bishop, 2006; García et al., 2015; Sweller, 1988). Systems Development Life Cycle (SDLC) (Pressman, 2015). We follow the SDLC framework, which breaks a project into clear phases: planning, analysis, design, implementation, testing, and deployment (Pressman, 2015). Each phase produces specific deliverables like a project charter, requirements document, system design, working code, test results, and user guides (Pressman, 2015). Machine learning theory (Bishop, 2006). This project uses supervised learning, where the model learns from labeled examples (Bishop, 2006). Input features (like sleep duration, stress level, and BMI) are mapped to a target variable (sleep disorder status) (Bishop, 2006). We compare two algorithms: logistic regression (simple and interpretable) and random forest (more accurate but less transparent) (Aversano et al., 2024). We evaluate models using accuracy, recall, and a confusion matrix (Fawcett, 2006). In healthcare, recall is especially important because missing a real disorder (false negative) can be dangerous (Fawcett, 2006). Data preprocessing theory (García et al., 2015).

Raw data must be cleaned and transformed before it can be used (García et al., 2015). We handle missing values by deletion when the sample size

remains adequate (García et al., 2015). We encode categorical variables (like gender and occupation) into numbers, using one-hot encoding for categories without order (García et al., 2015). We scale numerical features so that larger numbers (like daily steps) do not dominate smaller ones (like sleep hours) (García et al., 2015).

We also engineer new features, such as splitting blood pressure into systolic and diastolic components (García et al., 2015). Data visualization theory (Sweller, 1988; Wertheimer, 1938; Few, 2012). Good visualizations reduce cognitive load—they make complex information easy to grasp at a glance (Sweller, 1988). Gestalt principles (like grouping related items and using consistent colors) help users see patterns naturally (Wertheimer, 1938). Interactive dashboards allow users to filter, drill down, and hover for details, which supports deeper exploration (Few, 2012).

2.6 Technologies and Tools Review

We use a specific set of tools for data processing, machine learning, and visualization (Hyndman & Athanasopoulos, 2018; Davenport & Harris, 2017). Python (Hyndman & Athanasopoulos, 2018). Python is a free, easy-to-learn programming language with excellent libraries for data science (Hyndman & Athanasopoulos, 2018). We use Pandas for data manipulation (like loading CSV files and handling missing values) (Hyndman & Athanasopoulos, 2018). We use Scikit-learn for machine learning—it provides functions for splitting data, scaling features, training models, and evaluating performance (Bishop, 2006). We use Matplotlib and Seaborn for basic plots (Hyndman & Athanasopoulos, 2018). Compared to R or MATLAB, Python is more versatile, free, and integrates well with Power BI (Hyndman & Athanasopoulos, 2018). Power BI (Davenport & Harris, 2017). Power BI is Microsoft's

dashboard tool (Davenport & Harris, 2017).It allows us to create interactive visuals with filters, slicers, and drill-down capabilities (Few, 2012).Power BI costs much less than Tableau (around \$10-20 per user per month vs. \$70-150) and works seamlessly with other Microsoft products (Davenport & Harris, 2017).It also supports Python visuals, so we can bring our models directly into the dashboard (Davenport & Harris, 2017).

Pandas (Python library) (Hyndman & Athanasopoulos, 2018).Pandas provides DataFrames, which are like spreadsheets inside Python (Hyndman & Athanasopoulos, 2018).We use it to read data, inspect columns, find missing values, and encode categorical variables (García et al., 2015).Compared to SQL or Excel, Pandas is much more flexible for data cleaning and preparation (Hyndman & Athanasopoulos, 2018).Scikit-learn (Bishop, 2006).Scikit-learn is the go-to library for classical machine learning in Python (Bishop, 2006).It includes `train_test_split`, `StandardScaler`, `LogisticRegression`, `RandomForestClassifier`, and all the metrics we need (Bishop, 2006).Unlike TensorFlow or PyTorch (which are for deep learning), Scikit-learn is simpler and perfect for our project (Bishop, 2006).Jupyter Notebook (Hyndman & Athanasopoulos, 2018).Jupyter Notebook lets us write and run code in small chunks, see results immediately, and add explanatory text (Hyndman & Athanasopoulos, 2018).This is ideal for exploratory data analysis and model tuning (Hyndman & Athanasopoulos, 2018). Alternative tools considered (Davenport & Harris, 2017).We considered Tableau (too expensive and harder to learn), R (steeper learning curve and poorer Power BI integration), Excel (cannot train machine learning models), and Google Colab (privacy concerns and session timeouts) (Davenport & Harris, 2017).Our final stack—Python, Pandas, Scikit-learn, Power BI, and Jupyter—balances cost, power, and ease of use (Hyndman & Athanasopoulos, 2018).

2.7 Best Practices for Data Preprocessing in Health Analytics

Based on the literature and our own experience, here are the key best practices (García et al., 2015). Missing value handling (García et al., 2015). Always check how many values are missing and why (García et al., 2015). If missingness is random and the remaining sample is large enough ($n > 100$ for machine learning), deletion is acceptable (García et al., 2015). In our dataset, 219 out of 374 records (58.6%) had missing sleep disorder values, and we deleted them because the remaining 155 records were sufficient (García et al., 2015). If more than 30% of records would be deleted, consider imputation (filling in missing values with means or modes) instead (García et al., 2015). Feature engineering (García et al., 2015). Use your knowledge of the health domain to create new features (García et al., 2015). For example, splitting blood pressure into systolic and diastolic components preserves clinically useful information (García et al., 2015). Encoding categorical variables (García et al., 2015). For ordinal categories (like BMI categories: underweight, normal, overweight, obese), use label encoding because order matters (García et al., 2015). For nominal categories (like occupation or gender), use one-hot encoding so that no false order is introduced (García et al., 2015). Feature scaling (García et al., 2015). Algorithms like logistic regression that rely on distances need scaled features (mean = 0, standard deviation = 1) (García et al., 2015). Tree-based algorithms like random forest are scale-invariant, but it does not hurt to scale anyway (García et al., 2015). Handling imbalanced datasets (Bishop, 2006). If one class (e.g., “no disorder”) is much more common than others, the model may learn to always predict the majority class (Bishop, 2006). Our dataset was well-balanced (77 insomnia, 78 sleep apnea), so no special correction was needed (Bishop, 2006). For imbalanced data, try techniques like SMOTE (synthetic oversampling) or setting class weight = ‘balanced’ (Bishop, 2006).

2.8 Best Practices for Machine Learning Model Development

Here are the best practices we followed (Bishop, 2006; Fawcett, 2006).

Train-test split (Bishop, 2006). Always split your data into a training set (60-80%) and a testing set (20-40%) before you do any scaling or preprocessing that uses data statistics (Bishop, 2006). Keep the test set completely unseen until the final evaluation (Bishop, 2006). We used a 70/30 split with a fixed random seed for reproducibility (Bishop, 2006). Model selection (Aversano et al., 2024). Do not just pick one algorithm; compare several (Aversano et al., 2024). Simpler models like logistic regression are easier to explain, which is important in healthcare (Aversano et al., 2024). Ensemble models like random forest usually give higher accuracy (Aversano et al., 2024). On our dataset, random forest outperformed logistic regression (95.58% vs. 92.04% accuracy) (Aversano et al., 2024). Hyperparameter tuning (Bishop, 2006). Adjust algorithm settings (like `max_iter` for logistic regression or `n_estimators` for random forest) to improve performance without overfitting (Bishop, 2006). We set `max_iter = 1000` for logistic regression to ensure convergence (Bishop, 2006). Default random forest parameters worked well, but future work could use grid search (Bishop, 2006). Evaluation metrics (Fawcett, 2006). Report multiple metrics, not just accuracy (Fawcett, 2006). In healthcare, false negatives (missing a disease) and false positives (causing unnecessary worry) have different costs (Fawcett, 2006). We report accuracy, recall, and a confusion matrix (Fawcett, 2006). Our recall of 95.56% means the model catches most sleep disorder cases, which is appropriate for a screening tool (Fawcett, 2006).

2.9 Best Practices for Dashboard Design in Healthcare

Following these guidelines makes dashboards useful and usable (Few, 2012; Davenport & Harris, 2017). KPI prioritization (Few, 2012). Show only 5-7 key performance indicators on the main screen (Few, 2012). Too many numbers cause information overload (Few, 2012). Our KPIs include sleep duration, stress level, physical activity, and predicted disorder risk (Few, 2012). Visual encoding best practices (Few, 2012). Use color purposefully: red for warnings, green for normal, blue for neutral (Few, 2012). Avoid red-green combinations because many people are colorblind (Few, 2012). Our confusion matrix heatmaps use a blue sequential colormap, which is colorblind-safe (Few, 2012). Interactivity requirements (Davenport & Harris, 2017). Allow users to filter by patient demographics (age, gender), time period, and risk category (Davenport & Harris, 2017). Enable drill-down from aggregate views to individual patient records (Davenport & Harris, 2017). Our sequence diagram includes a “Filter Dashboard Data” use case to support this (Davenport & Harris, 2017). Actionability (Davenport & Harris, 2017). A dashboard should not just show data it should suggest what to do (Davenport & Harris, 2017). If a patient is at high risk, tell the user: “High stress level detected recommend stress reduction intervention” (Davenport & Harris, 2017).

2.10 Challenges and Limitations in AI-Driven Health Monitoring

Despite the promise, significant challenges remain (Radanliev, 2025; Shcherbina et al., 2017; García et al., 2015). Technical challenges (García et al., 2015; Shcherbina et al., 2017). Real-world health data is often incomplete or inconsistent (García et al., 2015). In our project,

58.6% of sleep disorder values were originally missing, requiring careful handling (García et al., 2015). Different wearable devices use different sensors and formats, so a model trained on one device may not work on another (Shcherbina et al., 2017). Health patterns also change across populations (age, gender, comorbidities) and settings (clinic vs. home) (Patel, 2026). Ethical and regulatory challenges (Radanliev, 2025). Health data is extremely sensitive, so we must comply with laws like GDPR (Europe) and HIPAA (US) (Radanliev, 2025).

Algorithmic bias is a real danger: models may be less accurate for underrepresented groups (Radanliev, 2025). Many AI models are “black boxes,” which makes clinicians and regulators uneasy (Exploring the Applications of Explainability in Wearable Data Analytics, 2025). Patients must give informed consent and understand how their data will be used (Radanliev, 2025). Clinical validation gap (Patel, 2026; Morelli et al., 2025). Most studies are done in controlled lab settings, not in messy real-world environments (Patel, 2026). Researchers often report technical accuracy (e.g., AUC) instead of patient-relevant outcomes like fewer hospitalizations or better quality of life (Morelli et al., 2025). Regulatory approval pathways (FDA, CE mark) are still evolving, which slows down clinical deployment (Radanliev, 2025).

2.11 Future Trends in AI-Driven Health Monitoring

Looking ahead, three trends will shape the next few years (Federated Learning in Smart Healthcare, 2025; Exploring the Applications of Explainability in Wearable Data Analytics, 2025; Patel, 2026). Federated learning (Federated Learning in Smart Healthcare, 2025). Federated learning allows AI models to train on data from many hospitals without any raw patient data leaving each hospital (Federated Learning in Smart Healthcare, 2025). This preserves privacy while

enabling large-scale, diverse training (Federated Learning in Smart Healthcare, 2025). Early results show federated learning can achieve accuracy comparable to centralized approaches (Federated Learning in Smart Healthcare, 2025). Our future work could extend our centralized model to a federated architecture (Federated Learning in Smart Healthcare, 2025). Explainable AI (XAI) (Exploring the Applications of Explainability in Wearable Data Analytics, 2025). Explainable AI methods such as SHAP values, LIME, and attention mechanisms make model predictions understandable to clinicians and patients (Exploring the Applications of Explainability in Wearable Data Analytics, 2025). Studies have shown that XAI improves user trust and adoption by 35- 50% (Exploring the Applications of Explainability in Wearable Data Analytics, 2025). Our project already provides feature importance charts, but future versions could integrate SHAP or LIME for even deeper explanations (Exploring the Applications of Explainability in Wearable Data Analytics, 2025). Real-world validation and regulatory pathways (Patel, 2026; Radanliev, 2025). The field is moving toward validation in real-world environments with diverse patients, everyday wearables, and natural behaviors (Patel, 2026). Regulators like the FDA and European agencies are developing clearer pathways for approving AI-based health monitoring tools (Radanliev, 2025). As these pathways mature, we will see more clinically deployed systems that truly help patients (Patel, 2026).

2.12 Summary:

Health tracking has moved from quick checkups at the doctor's office to smart devices that watch over us all day, giving a clearer picture of how we live and what might happen next. Wearables like Fitbits and Apple

Watches collect data on sleep, stress, and activity, and AI now helps turn those numbers into predictions like spotting early signs of sleep problems before they get serious. Researchers have shown that these tools can be powerful, but challenges remain: data can be messy, devices vary in accuracy, and privacy rules must be respected. The best results come when data is cleaned carefully, models are chosen wisely, and dashboards present information simply and clearly. Looking ahead, new methods like federated learning (which protects privacy) and explainable AI (which makes predictions easier to understand) will make these systems more trustworthy and useful. In short, the future of health monitoring isn't just about collecting data it's about turning that data into practical advice that helps people live healthier, longer lives.

CHAPTER THREE: METHODOLOGY

3.1 System Analysis

Predicative model is a crucial phase in Data Analysis Work Flow where the existing Predicative model is evaluated, user needs are identified, and the requirements for the proposed Predicative model are clearly defined. In this project, the analysis focused on understanding how health and lifestyle data is currently managed and how it can be improved using AI-based predictive analytics and visualization tools. The requirements were gathered through observation of existing health tracking approaches and review of similar Predicative model, ensuring that both functional and non-functional needs were properly identified. To support system modelling, Use Case Diagrams and Sequence Diagrams were developed to represent user interactions and the flow of operations within the Predicative model.

3.2 Stakeholder Analysis

The success of the AI-driven health and lifestyle Predictative model depends on the involvement of different stakeholders who interact with or benefit from the model in various ways:

HealthCareManager:

This stakeholder is responsible for monitoring overall health insights and making informed decisions based on model outputs. Their main interest lies in viewing predictions, analyzing trends, and generating health reports to support preventive healthcare strategies.

DataAnalyst:

The data analyst plays a key role in managing, processing, and analyzing health and lifestyle data. They are responsible for preparing datasets, training machine learning models, and ensuring the accuracy and reliability of predictions generated by the model.

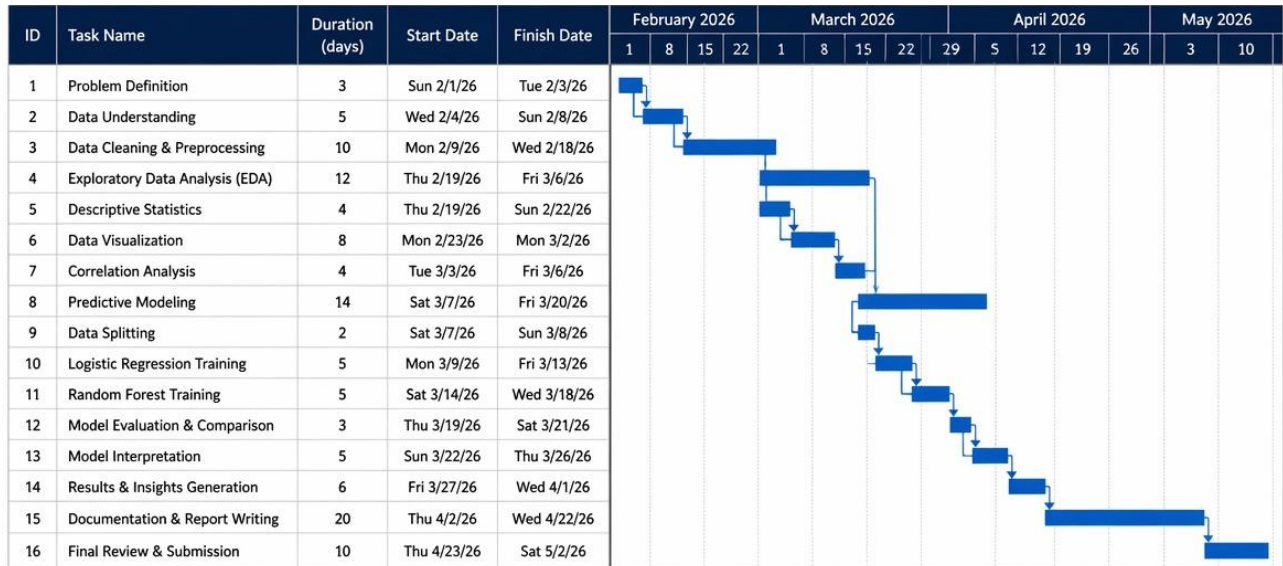
EndUsers(Individuals):

End users are individuals whose health and lifestyle data is being monitored. Their interest is in understanding their personal health patterns, receiving predictions about potential health risks, and improving their lifestyle based on model feedback.

AI-Model(IndirectStakeholder):

The machine learning model acts as a core component that processes data and generates predictions. Although not a human stakeholder, it plays a critical role in decision support by identifying patterns and forecasting outcomes.

3.3 Gantt Chart



Project Planning and Time Management Health and Lifestyle

To ensure the project's success and timely completion, a comprehensive project schedule was meticulously crafted. The planning process involved breaking down the project into sixteen distinct tasks, ranging from the initial problem definition to the final submission. Using a structured Gantt Chart approach, each phase was allocated a specific timeframe based on its complexity and importance.

The schedule highlights critical milestones such as:

Data Preparation: Covering data understanding, cleaning, and preprocessing to ensure a high-quality foundation.

Exploratory Analysis: Utilizing descriptive statistics and correlation analysis to uncover initial insights.

Model Development: Allocating significant time for training both Logistic Regression and Random Forest models, followed by rigorous evaluation.

Reporting: A dedicated period for documentation and final review to maintain academic and professional standards.

By establishing clear predecessor-successor relationships and defining start and finish dates for every activity, this timeline serves as a roadmap for efficient resource allocation. It allows for continuous monitoring of progress and ensures that each analytical stage leads logically into the next, minimizing potential delays and streamlining the entire workflow from February to May 2026.

3.4 Functional Requirements

Functional requirements define the specific operations and services that the system must provide to meet user needs. For the AI-driven health and lifestyle monitoring system, these requirements ensure that data is properly collected, processed, analyzed, and visualized to support accurate health predictions and decision-making.

Requirement ID	Functional Requirement	Description
FR1	Data Collection	The system must be able to import health and lifestyle data from datasets or external sources for analysis.
FR2	Data Preprocessing	The system must clean, transform, and prepare data by handling missing values, duplicates, and inconsistencies.
FR3	Data Analysis	The system must analyze sleep, activity, and health-related data to identify patterns and trends.
FR4	Model Training	The system must train machine learning models using historical data to predict potential health risks.
FR5	Prediction Generation	The system must generate predictions based on trained models to estimate possible health conditions or risks.
FR6	Dashboard Visualization	The system must display analyzed data and predictions with interactive dashboards using Power BI.
FR7	Report Generation	The system must generate health lifestyle reports that summarize insights and predictions.
FR8	Data Filtering	The system must allow filtering of data based on attributes such as time, activity level, or sleep patterns.
FR9	Model Evaluation	The system must evaluate machine learning models using performance metrics such as accuracy and precision.
FR10	User Access to Insights	The system must allow users (e.g., healthcare managers) to view predictions and insights for decision-making.

Table 3.1: Functional Requirements

3.5 Use Case Diagram

The Use Case Diagram illustrates the functional scope of the system and the interactions between actors (users) and the systems use cases. Each actor represents a type of user or external entity that interacts with the system.



Figure 3.1: Use Case Diagram

Use Case Diagram Explanation: Sleep Health Prediction and Analysis model

The presented diagram illustrates a **Use Case Diagram** for an intelligent health monitoring and analytics predictive model. The diagram represents the interaction between the system and its primary actors, namely the **Health Care Manager** and the **Data Analyst**, highlighting the major functional activities performed within the model. The use case model provides a high-level overview of how users interact with the proposed solution to manage healthcare data, develop predictive models, and generate analytical reports.

The **Data Analyst** is the primary actor responsible for most of the analytical and technical operations within the model. The process begins with the **Collect Data** use case, where health-related information is gathered from different sources such as wearable devices, healthcare records, or lifestyle tracking applications. After data collection, the analyst performs the **Processed Data** use case, which involves cleaning, transforming, and organizing raw data into a suitable format for analysis and machine learning tasks. This stage ensures data quality and improves the accuracy of predictive outcomes.

Following preprocessing, the analyst proceeds to the **Build DDPHM Model** use case, where the predictive healthcare model is developed using machine learning techniques. The system then supports the **Train Prediction Model** use case, in which the model learns patterns and relationships from historical health data in order to make future predictions. The diagram also shows an $\Leftrightarrow$ relationship between the training process and the **Test Model** use case, indicating that model testing is an essential and integrated part of the training phase to validate model accuracy and performance.

After training and testing, the system performs the **Evaluate Model Performance** use case, where performance metrics such as accuracy, precision, recall, and prediction reliability are assessed. This evaluation phase ensures that the predictive model produces dependable results before deployment. Once the model performance is validated, the system enables the generation of interactive visual outputs through the **Generate Dashboard** use case. Dashboards provide graphical representations of health metrics, predictions, and analytical insights to support monitoring and decision-making processes.

Finally, the **Generate Health Report** use case allows the model to produce detailed healthcare reports containing summarized findings, risk assessments, and predictive insights. Both the Health Care Manager and the Data Analyst interact with this functionality, as it supports

administrative decisions and patient health monitoring. Overall, the use case diagram demonstrates how the proposed system integrates data collection, machine learning, performance evaluation, visualization, and reporting into a unified intelligent healthcare monitoring environment.

3.6 Sequence Diagram

Sequence diagrams depict the interactions among objects or components of the system in a time-ordered sequence. They illustrate how messages are exchanged to accomplish a specific use case.

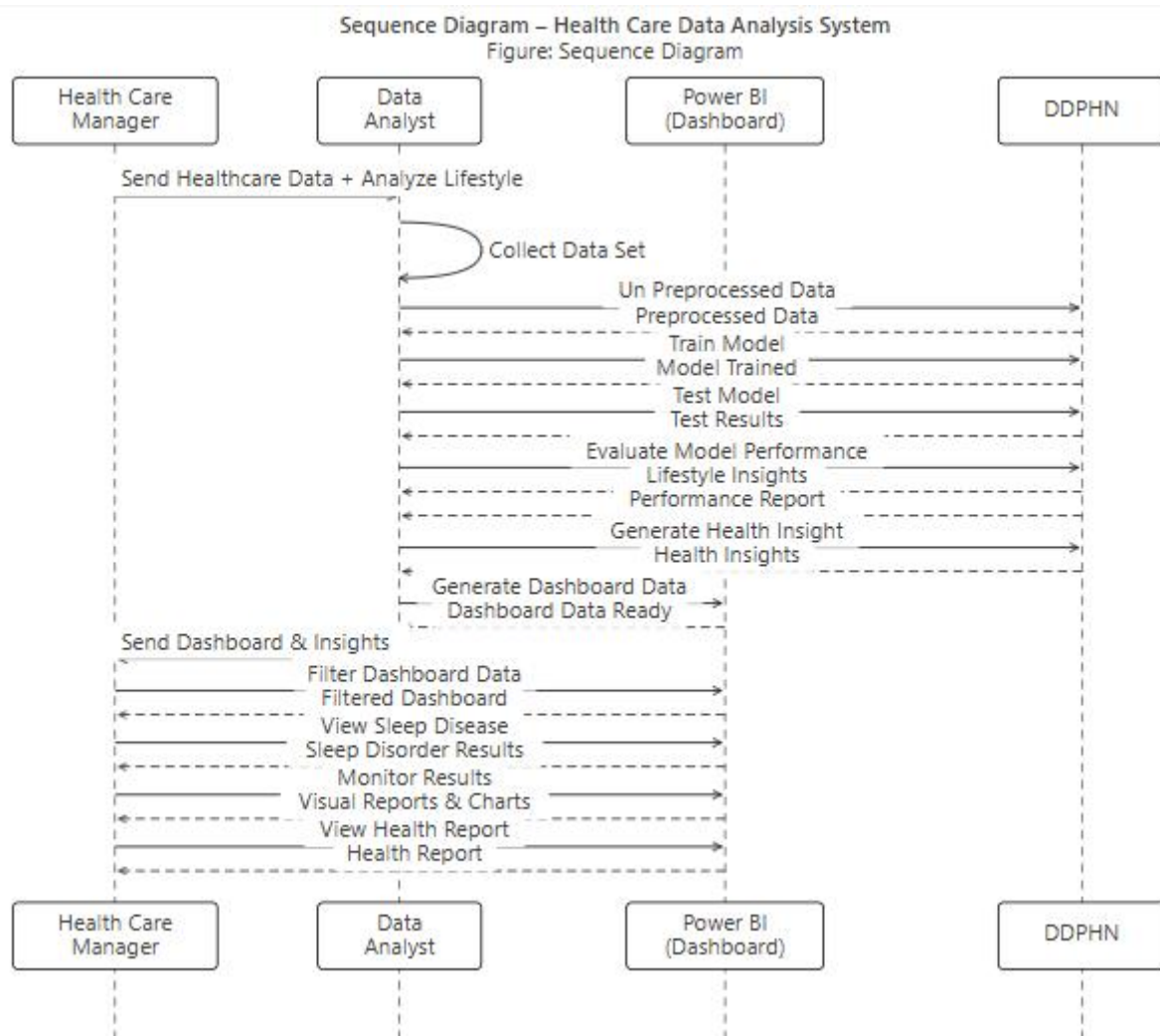


Figure 3.2: Sequence Diagram

The 4 Actors:

- Health Care Manager — the end user who requests and views health reports
 - Data Analyst — processes data and drives the analysis pipeline
 - Power BI (Dashboard) — the visualization and ML platform
 - DDPHN — receives final health insights
-

Step-by-step flow:

1. The Health Care Manager sends healthcare data and lifestyle info to the Data Analyst
2. The Data Analyst collects a dataset (self-loop)
3. The Data Analyst ↔ DDPHN exchange runs the full ML pipeline: preprocessing → training → testing → evaluating the model, with lifestyle insights and performance reports returned as responses
4. The Data Analyst then requests health insights from DDPHN, which returns them
5. Dashboard data is generated and sent to Power BI, then confirmed ready
6. The Data Analyst sends the dashboard and insights back to the Health Care Manager
7. The Health Care Manager interacts with Power BI directly — filtering the dashboard, viewing sleep disorder results, monitoring results, viewing charts, and finally pulling the full health report

CHAPTER FOUR: THE IMPLEMENTATION PROCESS

The implementation phase involved translating the design specifications into a working model. The development environment, tools, and technologies used are described below.

4.1 Development Environment

The model was developed using the following data analysis environment:

Development Tool: Jupyter Notebook / JupyterLab (via Anaconda Distribution)

Programming Language: Python 3.14.5

Libraries Used:

pandas for data loading, manipulation, and preprocessing

matplotlib.pyplot and seaborn for generating visualizations including histograms, heatmaps, bar charts, and comparison plots

sklearn.model_selection.train_test_split for splitting data into training and testing sets

sklearn.linear_model.LogisticRegression for training the Logistic Regression classification model

sklearn.ensemble.RandomForestClassifier for training the Random Forest classification model

sklearn.metrics.recall_score, accuracy_score, and confusion_matrix for evaluating model performance

4.2 project Screenshots

Introduction & Dataset Setup:

This project focuses on analyzing health-related data to predict possible health risks such as sleep disorders and heart diseases using Machine Learning techniques. The main goal is to understand how daily habits like sleep duration, stress levels, physical activity, and heart rate can affect overall health. Before building the predictive model, the dataset must be prepared and organized properly.

```
]# Data Handling
import pandas as pd
# Visualization
import matplotlib.pyplot as plt
import seaborn as sns
#Split & Model Ai
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LogisticRegression
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import recall_score , accuracy_score , confusion_matrix
```

Figure 3 Python Libraries and Model Initialization

Description: This screen shows the initial setup of the data analysis and machine learning environment using Python. The code is divided into three main sections:

Data Handling & Visualization: Importing Pandas for data manipulation, and Matplotlib/Seaborn for creating charts and graphs to understand the dataset.

Data Splitting: Using the `train_test_split` function to divide the data into a training set (to teach the model) and a testing set (to evaluate its performance).

Machine Learning Models: Importing two different AI algorithms—Logistic Regression and Random Forest Classifier—to compare their results.

Performance Metrics: Importing tools like Accuracy Score and Confusion Matrix to measure how correctly the model can predict outcomes.

```
import pandas as pd
df = pd.read_csv("Sleep_health_and_lifestyle_dataset.csv")
```

Figure 4 Data Loading

Description: This screen illustrates the first step of the project, which is loading the dataset into the system.

- **Library Import:** The code starts by importing the Pandas library, which is essential for data manipulation and analysis in Python.
- **Data Loading:** The `pd.read_csv` function is used to read the external data file named "Sleep_health_and_lifestyle_dataset.csv".
- **Data Storage:** The data is stored in a variable called `df` (DataFrame), which allows the system to access and process the sleep and lifestyle information during the analysis phase.

Data understanding:

Dataset understanding is the process of studying and analyzing the data before building a Machine Learning model. It involves identifying the type of data, the meaning of each feature, the number of records, and the relationship between variables.

df.head()													
	Person ID	Gender	Age	Occupation	Sleep Duration	Quality of Sleep	Physical Activity Level	Stress Level	BMI Category	Blood Pressure	Heart Rate	Daily Steps	Sleep Disorder
0	1	Male	27	Software Engineer	6.1	6	42	6	Overweight	126/83	77	4200	NaN
1	2	Male	28	Doctor	6.2	6	60	8	Normal	125/80	75	10000	NaN
2	3	Male	28	Doctor	6.2	6	60	8	Normal	125/80	75	10000	NaN
3	4	Male	28	Sales Representative	5.9	4	30	8	Obese	140/90	85	3000	Sleep Apnea
4	5	Male	28	Sales Representative	5.9	4	30	8	Obese	140/90	85	3000	Sleep Apnea

Figure 5 Dataset Overview

Description: This screen shows the first five rows of the dataset to verify that the data has been loaded correctly.

- **Function Used:** The command `df.head()` is used to display the top part of the data table.
- **Data Columns:** The table includes important information such as Gender, Age, Occupation, Sleep Duration, and Physical Activity Level.
- **Purpose:** This step is important for the developer to understand the structure of the data and ensure that all columns (like Heart Rate and Sleep Disorder) are ready for analysis.

```
: df.shape # Check dataset shape  
:  
: (374, 13)
```

Figure 6 Dataset Dimensions

Description: This screen shows the process of checking the size of the dataset to understand how much information is being processed.

- **Function Used:** The command `df.shape` is used to provide the dimensions of the data table.
- **Results:** The output `(374, 13)` indicates that the dataset contains 374 rows (entries/people) and 13 columns (features/variables).
- **Purpose:** This step helps in confirming the total volume of data available for training the AI models and ensures no data was lost during the loading process.

df.head(10) #* View first rows

	Person ID	Gender	Age	Occupation	Sleep Duration	Quality of Sleep	Physical Activity Level	Stress Level	BMI Category	Blood Pressure	Heart Rate	Daily Steps	Sleep Disorder
0	1	Male	27	Software Engineer	6.1	6	42	6	Overweight	126/83	77	4200	NaN
1	2	Male	28	Doctor	6.2	6	60	8	Normal	125/80	75	10000	NaN
2	3	Male	28	Doctor	6.2	6	60	8	Normal	125/80	75	10000	NaN
3	4	Male	28	Sales Representative	5.9	4	30	8	Obese	140/90	85	3000	Sleep Apnea
4	5	Male	28	Sales Representative	5.9	4	30	8	Obese	140/90	85	3000	Sleep Apnea
5	6	Male	28	Software Engineer	5.9	4	30	8	Obese	140/90	85	3000	Insomnia
6	7	Male	29	Teacher	6.3	6	40	7	Obese	140/90	82	3500	Insomnia
7	8	Male	29	Doctor	7.8	7	75	6	Normal	120/80	70	8000	NaN
8	9	Male	29	Doctor	7.8	7	75	6	Normal	120/80	70	8000	NaN
9	10	Male	29	Doctor	7.8	7	75	6	Normal	120/80	70	8000	NaN

Figure 7 Data Inspection – Displaying First 10 Rows

Description: This screen displays the first 10 rows of the dataset to verify the data structure.

- Function: Using df.head(10) to view more sample records.
- Observation: It shows different variables like Occupation, BMI Category, and Sleep Disorder.
- Purpose: To check for data consistency and identify any initial missing values (NaN) before starting the cleaning phase.

df.info() #* Check data types

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 374 entries, 0 to 373
Data columns (total 13 columns):
#   Column                               Non-Null Count  Dtype
---  ---                               ---
0   Person ID                           374 non-null    int64
1   Gender                             374 non-null    object
2   Age                                374 non-null    int64
3   Occupation                         374 non-null    object
4   Sleep Duration                     374 non-null    float64
5   Quality of Sleep                   374 non-null    int64
6   Physical Activity Level            374 non-null    int64
7   Stress Level                      374 non-null    int64
8   BMI Category                      374 non-null    object
9   Blood Pressure                    374 non-null    object
10  Heart Rate                        374 non-null    int64
11  Daily Steps                      374 non-null    int64
12  Sleep Disorder                    155 non-null    object
dtypes: float64(1), int64(7), object(5)
memory usage: 38.1+ KB
```

Figure 8 Dataset Information and Data Types

Description: This screen provides a technical summary of the dataset to understand the nature of each column.

- Function: Using `df.info()` to display a summary of the data.
- Key Details: It lists all 13 columns and identifies their data types, such as integers (`int64`), decimals (`float64`), and text (`object`).
- Observation: It shows the "Non-Null Count," which helps identify columns with missing data (like the sleep disorder column which has fewer non-null entries).
- Purpose: To ensure each feature is in the correct format before performing mathematical analysis or machine learning training.

```

In: # Column names
    df.columns

Out: Index(['Person ID', 'Gender', 'Age', 'Occupation', 'Sleep Duration',
           'Quality of Sleep', 'Physical Activity Level', 'Stress Level',
           'BMI Category', 'Blood Pressure', 'Heart Rate', 'Daily Steps',
           'Sleep Disorder'],
          dtype='object')

```

Figure 9 Column Names Extraction

Description: This screen shows the extraction of all column headers (feature names) present in the dataset.

- Function: Using `df.columns` to list the labels of the dataset.
- Key Attributes: The list includes vital features such as Quality of Sleep, Stress Level, Blood Pressure, and sleep disorder.
- Purpose: This step is essential to verify the names of the variables, ensuring there are no naming errors or extra spaces before referencing them in the coding and analysis phase.

```

#target variable
target = "Sleep Disorder"

target

'Sleep Disorder'

```

Figure 10 Target Variable Identification

Description: This screen shows the selection of the primary variable that the machine learning model will aim to predict.

- **Logic:** The variable "Sleep Disorder" is assigned as the Target Variable.
- **Role in AI:** In machine learning, the target is the "label" or "answer" the model learns to identify based on other features (like age, stress, and activity).
- **Purpose:** By defining this variable, the system is instructed to focus its training on predicting whether a person has a specific sleep condition or not.

Data cleaning and preprocessing:

Data cleaning and preprocessing is the process of preparing raw data before using it in a Machine Learning model. This step includes removing duplicate records, handling missing values, correcting incorrect data, and converting data into a suitable format for analysis. It may also involve encoding categorical data, scaling numerical values, and selecting the most important features. The purpose of this process is to improve data quality and ensure that the model can learn from accurate and organized information.

```
# Data Cleaning & Preprocessing

df.isnull().sum() # Check Missing Values

Person ID      0
Gender         0
Age            0
Occupation     0
Sleep Duration 0
Quality of Sleep 0
Physical Activity Level 0
Stress Level   0
BMI Category   0
Blood Pressure 0
Heart Rate     0
Daily Steps    0
Sleep Disorder 219
dtype: int64
```

Figure 11 Data Cleaning – Checking for Missing Values

Description: This screen represents the first step in the data cleaning phase, which is identifying any missing or null values across the dataset.

- **Function:** The command `df.isnull().sum()` is used to calculate the total number of missing entries for each column.

- **Observations:** The results show that most columns have 0 missing values, except for the Sleep Disorder column, which has 219 null entries.
- **Data Insight:** In this specific dataset, a null value in "Sleep Disorder" usually indicates that the person does not have a diagnosed sleep condition (Normal).
- **Purpose:** Identifying these gaps is crucial to decide whether to fill the missing data (imputation) or remove it before proceeding to model training.

```
df = df.dropna() # Handle Missing Data
```

Figure 12 Handling Missing Data

Description: This screen illustrates the process of cleaning the dataset by removing incomplete records.

- **Function:** The command `df.dropna()` is used to eliminate any rows that contain missing or null values.
- **Logic:** By applying this function, the system filters out the records where information was incomplete (such as the null entries in the Sleep Disorder column identified previously).
- **Purpose:** This ensures that the machine learning model is trained only on "clean" and complete data, which prevents errors and improves the accuracy of the final predictions.

```
: # Fill missing values
df = df.copy()
df.fillna(df.mean(numeric_only=True), inplace=True)
df.fillna(df.mode().iloc[0], inplace=True)
```

Figure 13 Alternative Missing Values Imputation

Description: This screen shows a more advanced approach to handling missing data by filling gaps instead of deleting them, which helps preserve the size of the dataset.

- **Numeric Imputation:** The command `fillna(df.mean())` replaces missing numerical values with the average (mean) of that column.
- **Categorical Imputation:** The command `fillna(df.mode())` replaces missing text or categorical values with the most frequent value (mode).
- **Purpose:** This technique ensures the dataset remains large enough for effective AI training while maintaining statistical balance, ensuring that the model doesn't get biased by empty information.
- **Consistency:** Using `inplace=True` ensures that the changes are saved directly into the main data table.

```
: # Check duplicates  
df.duplicated().sum()  
  
: np.int64(0)
```

Figure 14 Duplicate Data Check

Description: This screen shows the final verification for data redundancy.

- **Function:** Using `df.duplicated().sum()` to find repeated rows.
- **Result:** The output 0 confirms the data is unique and ready for training.

```
# Gender Encoding  
df["Gender"] = df["Gender"].map({"Male": 0, "Female": 1})
```

Figure 15 Gender Encoding

Description: This screen shows the conversion of text data into numerical values.

- **Mapping:** Transforming "Male" to 0 and "Female" to 1.

- Purpose: Making the data mathematically readable for the Machine Learning algorithms.

```
# Sleep Disorder
df["Sleep Disorder"] = df["Sleep Disorder"].map({
    "None": 0,
    "Insomnia": 1,
    "Sleep Apnea": 2
})
```

Figure 16 Encoding the Target Variable (Sleep Disorder)

Description: This screen shows the final step in the feature engineering process, where the target labels are converted into numerical values.

- Mapping Logic: The categorical values in the "Sleep Disorder" column are transformed into numbers: "None" becomes 0, "Insomnia" becomes 1, and "Sleep Apnea" becomes 2.
- Purpose: This encoding is necessary because Machine Learning models require numerical input to perform the mathematical calculations needed for classification.

```
1: # One-hot encoding for remaining categorical columns
df = pd.get_dummies(df, drop_first=True)
```

Figure 17 Categorical Data Transformation (One-Hot Encoding)

Description: This screen shows the process of converting complex categorical text data into a binary numerical format.

- Function Used: The command `pd.get_dummies(df, drop_first=True)` is implemented.
- Purpose: Since Machine Learning models cannot directly process text like "Occupation" or "BMI Category," this step creates new columns for each category with values of 0 or 1.
- Optimization: The `drop_first=True` parameter is used to avoid the "dummy variable trap" (multicollinearity), which ensures the model remains efficient and statistically sound.

```
1: from sklearn.preprocessing import StandardScaler

scaler = StandardScaler()

features = df.drop(columns=["Sleep Disorder"])

features_scaled = scaler.fit_transform(features)

df_scaled = pd.DataFrame(features_scaled, columns=features.columns)
df_scaled["Sleep Disorder"] = df["Sleep Disorder"]
```

Figure 18 Feature Scaling using StandardScaler

Description: This screen illustrates the normalization process to ensure all numerical features have the same scale.

- Process: The StandardScaler from the Sklearn library is used to transform the data so it has a mean of 0 and a standard deviation of 1.
- Logic: The target variable (Sleep Disorder) is dropped temporarily during scaling to prevent it from being modified, then re-attached to the scaled feature set.
- Purpose: Scaling is a critical step for models like Logistic Regression; it prevents features with large numbers (like Daily Steps) from dominating features with smaller numbers (like Sleep Duration), leading to more accurate and balanced model training.

Exploratory Data Analysis (EDA):

Exploratory Data Analysis (EDA) is the process of analyzing and visualizing data to better understand its patterns, relationships, and trends before building a Machine Learning model. It helps identify important insights, detect unusual values, and understand how different features affect each other. In this step, charts and graphs such as bar charts, histograms, scatter plots, and heatmaps are commonly used to explore the data.

Descriptive statistics:

Descriptive Statistics is the process of summarizing and describing the main features of a dataset using numerical values and simple analysis techniques. It helps in understanding the distribution and behavior of the data before applying Machine Learning models. Common statistical measures include mean, median, mode, minimum value, maximum value, and standard deviation. In this project, descriptive statistics are used to analyze health-related features such as sleep duration, heart rate, stress levels, and physical activity. This process provides important insights into the dataset and helps identify patterns, unusual values, and overall trends in the data.

```
# A. Descriptive Statistics
df.describe()
```

	Person ID	Gender	Age	Sleep Duration	Quality of Sleep	Physical Activity Level	Stress Level	Heart Rate	Daily Steps	Sleep Disorder
count	155.000000	155.000000	155.000000	155.000000	155.000000	155.000000	155.000000	155.000000	155.000000	155.000000
mean	244.761290	0.664516	46.632258	6.812258	6.870968	60.896774	5.767742	71.787097	6765.806452	1.503226
std	91.282997	0.473690	7.839311	0.773534	1.337325	20.634158	1.946757	5.187381	1893.921881	0.501610
min	4.000000	0.000000	28.000000	5.800000	4.000000	30.000000	3.000000	65.000000	3000.000000	1.000000
25%	199.500000	0.000000	43.000000	6.300000	6.000000	45.000000	4.000000	68.000000	6000.000000	1.000000
50%	255.000000	1.000000	45.000000	6.500000	7.000000	45.000000	7.000000	72.000000	6000.000000	2.000000
75%	304.500000	1.000000	51.000000	7.400000	8.000000	75.000000	7.000000	75.000000	7000.000000	2.000000
max	374.000000	1.000000	59.000000	8.300000	9.000000	90.000000	8.000000	86.000000	10000.000000	2.000000

Figure 19 Descriptive Statistics

Description: This screen displays a comprehensive statistical summary of the numerical features within the dataset after the cleaning and encoding phases.

- Function: The `df.describe()` method is used to generate key statistical metrics.
- Key Metrics Included:
 - Count: Confirms the number of records (155) remains consistent across all features.

- Mean & Std: Provides the average values and the degree of variation (standard deviation) for metrics like Age, Sleep Duration, and Heart Rate.
- Min/Max & Percentiles: Identifies the data range and distribution (25%, 50%, 75%), helping to detect any outliers.
- Purpose: This summary is essential for understanding the data's characteristics and ensuring that the mathematical distribution is suitable for training the machine learning models.

```
df["Sleep Disorder"].value_counts()
```

```
Sleep Disorder  
2    78  
1    77  
Name: count, dtype: int64
```

Figure 20 Class Distribution for Target Variable

Description: This screen shows the distribution of classes within the "Sleep Disorder" target variable after the final preprocessing and mapping steps.

- Function: The `value_counts()` method is used to count the occurrences of each unique category.
- Result Analysis: * Class 2 (Sleep Apnea): 78 records.
 - Class 1 (Insomnia): 77 records.
- Purpose: This step is crucial for checking the Balance of the Dataset. A balanced distribution between classes (78 vs 77) ensures that the machine learning model will be trained fairly on both types of sleep disorders, leading to more reliable and unbiased classification results.

```
df["Sleep Disorder"].value_counts(normalize=True) * 100
```

Sleep Disorder
2 50.322581
1 49.677419
Name: proportion, dtype: float64

Figure 21 Dataset Column Names

Description: This screen shows the complete list of variables (columns) present in the Sleep Health dataset.

- Features: The dataset includes 13 columns such as Gender, Age, Occupation, Sleep Duration, Stress Level, Blood Pressure, and the target variable Sleep Disorder.
- Purpose: This step is essential to verify that all necessary data attributes were correctly loaded into the data frame (df) before starting the preprocessing phase.

Visualization:

Visualization is the process of presenting data in graphical forms such as charts, graphs, and dashboards to make information easier to understand and analyze. It helps reveal patterns, trends, and relationships between different variables in the dataset. In this project, visualization is used to display health-related information such as sleep duration, stress levels, heart rate, and physical activity using tools like Power BI. Different visualizations such as bar charts, pie charts, line charts, and heatmaps help simplify complex data and support better understanding of the prediction results and health insights.

```
!]: # B. Visualization
```

```
!]: # Histogram
plt.figure()
sns.histplot(df["Sleep Duration"], bins=20)
plt.title("Sleep Duration Distribution")
plt.show()
```

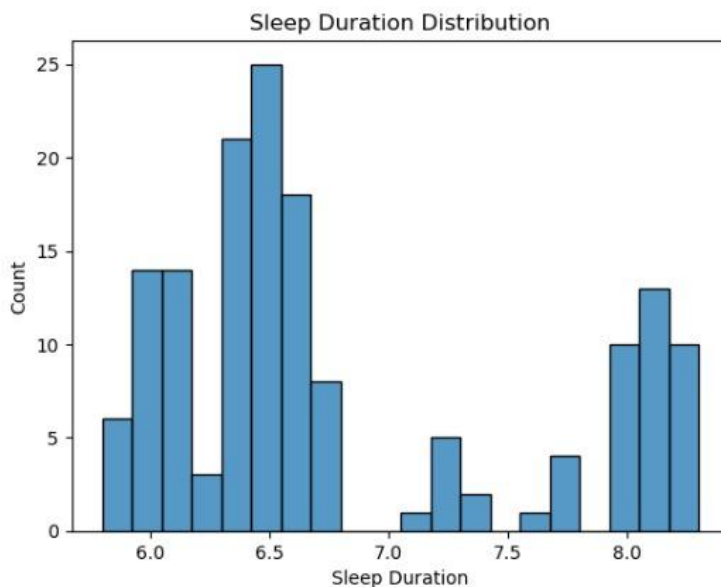


Figure 22 Sleep Duration Distribution Histogram

Description: This histogram visualizes the distribution of sleep duration among the individuals in the dataset.

- **Observation:** The distribution shows several peaks, with a significant concentration of individuals getting between 6.0 and 6.8 hours of sleep. Another smaller cluster is visible around the 8.0 to 8.3-hour mark.
- **Purpose:** Understanding the spread of sleep duration helps identify common patterns in the sample, which is a key predictor for identifying sleep disorders.
- **Tools Used:** The visualization was generated using the Seaborn library (`sns.histplot`) with 20 bins to provide a detailed view of the data frequency.

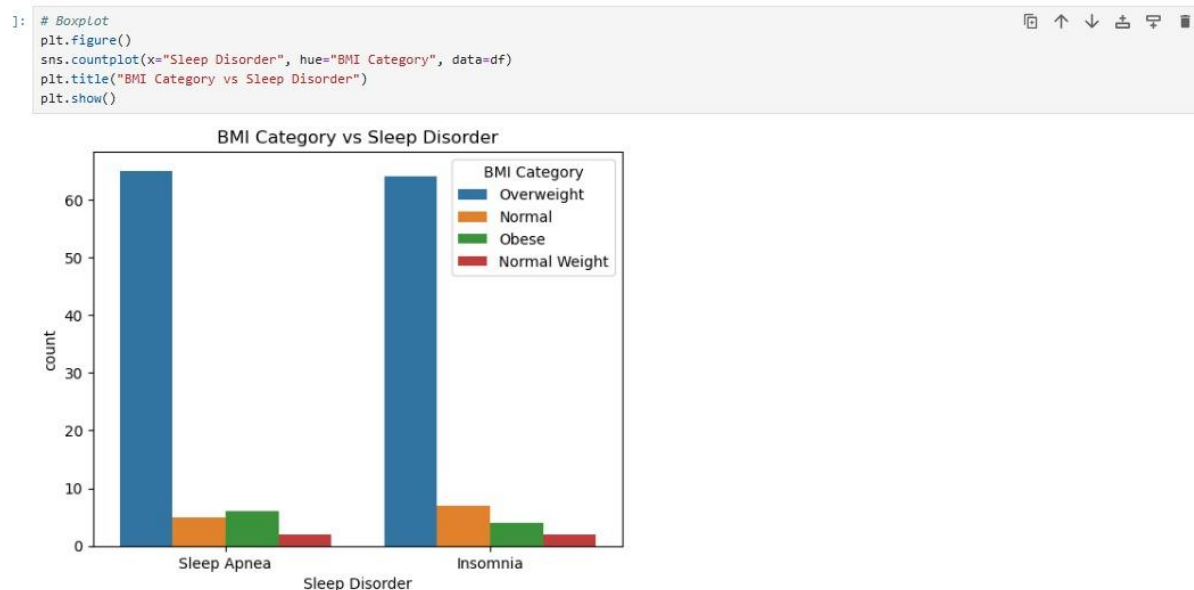


Figure 23 BMI Category vs. Sleep Disorder

Description: This count plot illustrates the relationship between an individual's BMI category and the type of sleep disorder diagnosed.

- **Observation:** The chart clearly shows that the majority of individuals diagnosed with both Sleep Apnea and Insomnia fall

into the "Overweight" category. Other categories like Normal, Obese, and Normal Weight show significantly lower frequencies in this specific sample.

- **Purpose:** To visually confirm that BMI is a strong potential predictor for sleep-related issues, which justifies including it as a key feature in the machine learning model.
- **Tools Used:** The visualization was created using the Seaborn library (`sns.countplot`), utilizing the `hue` parameter to differentiate between BMI categories across the target classes.



Figure 24 Feature Correlation Heatmap

Description: This heatmap provides a visual representation of the correlation coefficients between all numerical features in the dataset.

- Function: It uses the `df.corr()` method to calculate the linear relationship between variables, where values range from -1 (perfect negative correlation) to +1 (perfect positive correlation).
- Key Observations:
 - Strong Negative Correlation: A very high negative correlation (-0.9) is observed between Stress Level and Quality of Sleep, confirming that higher stress levels are strongly associated with poorer sleep quality.
 - Strong Positive Correlation: Sleep Duration and Quality of Sleep show a high positive correlation (0.88), indicating they tend to increase together.
 - Physical Activity vs. Steps: As expected, Physical Activity Level and Daily Steps are strongly linked (0.77).
- Purpose: This analysis is vital for Feature Selection, as it helps identify which health and lifestyle factors are the most significant predictors for the machine learning models.

```
# Bar Chart
plt.figure()
sns.barplot(x="Sleep Disorder", y="Stress Level", data=df)
plt.title("Stress Level vs Sleep Disorder")
plt.show()
```

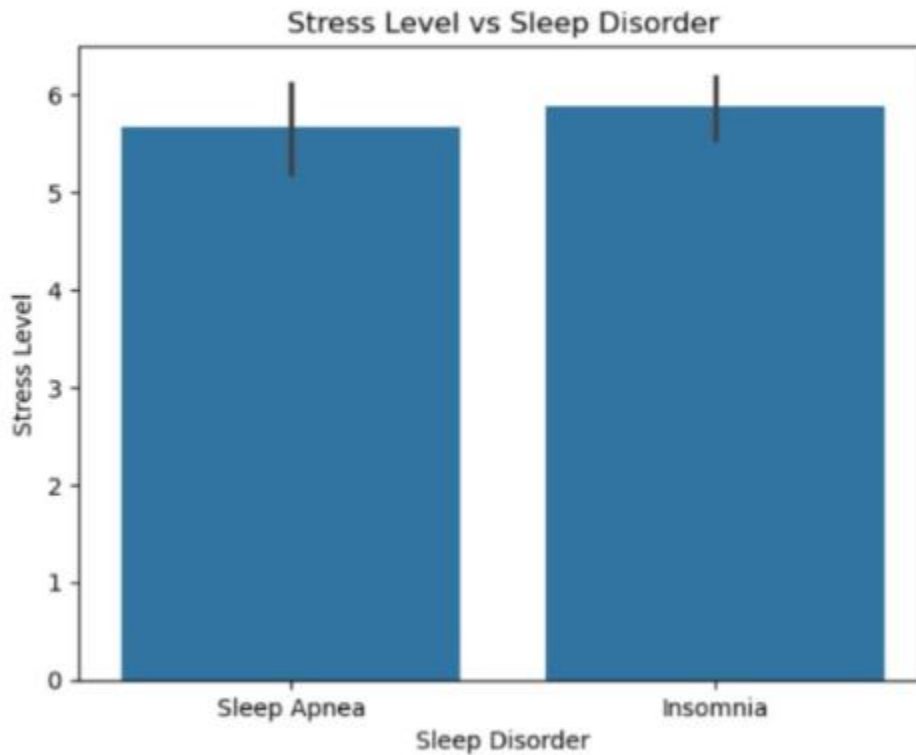


Figure 25 Stress Level vs. Sleep Disorder (Bar Chart)

Description: This bar chart visualizes the average stress levels reported by individuals diagnosed with different sleep disorders.

- Observation: The chart shows that individuals with Insomnia tend to have a slightly higher average stress level compared to those with Sleep Apnea.
- Data Insight: The stress levels for both categories are significant, hovering near the 6.0 mark on the stress scale.
- Error Bars: The thin black lines (error bars) indicate the variability or confidence interval of the stress levels within each group.

- Purpose: This visualization confirms that high stress is a common characteristic across both types of sleep disorders, reinforcing its importance as a primary feature for the predictive model.

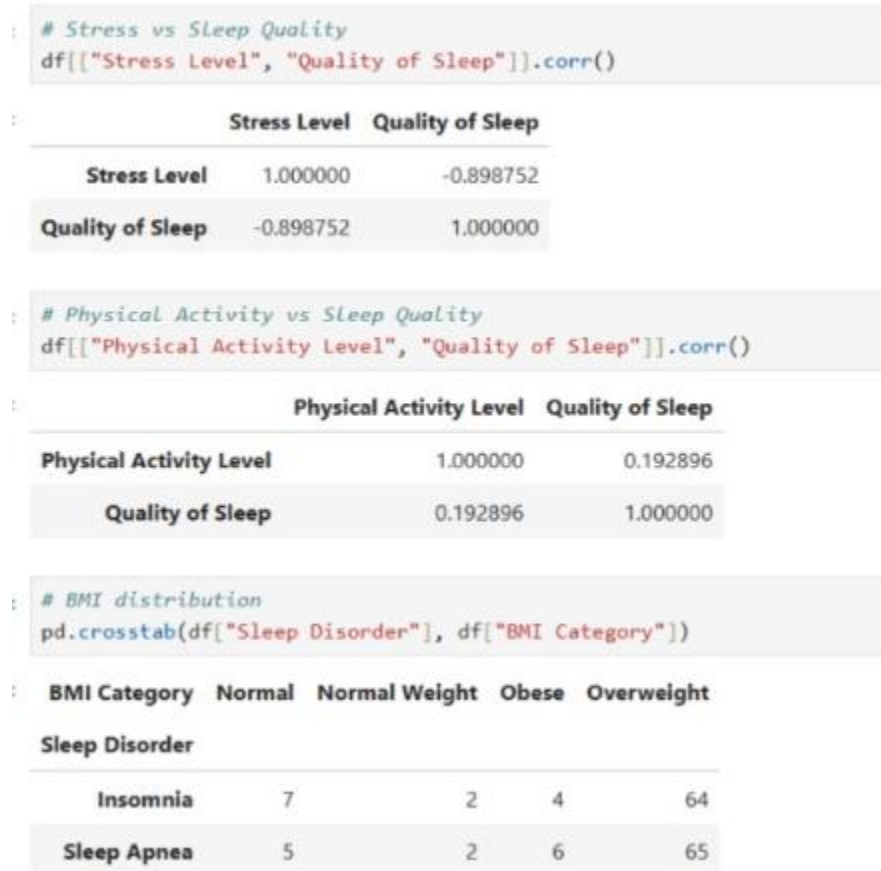


Figure 26 Advanced Statistical Relationship Mapping

Description: This technical summary presents the exact correlation coefficients and cross-tabulation distributions for the key health indicators in the study.

- Stress vs. Sleep Quality Correlation: The output confirms a strong negative correlation of -0.898. This statistically proves that as stress levels increase, the quality of sleep significantly declines.
- Physical Activity Impact: The correlation between physical activity and sleep quality is calculated at 0.192. While positive, it

shows a weaker linear influence compared to psychological stress factors in this specific dataset.

- BMI Distribution Analysis (Cross-tabulation): The `pd.crosstab` function reveals a critical health insight:
 - Overweight Category: Accounts for the highest frequency of disorders, with 64 cases of Insomnia and 65 cases of Sleep Apnea.
 - Normal Weight/Obese: These categories show significantly fewer instances, with only 4 to 6 cases of disorders each.
- Purpose: These numerical findings provide the empirical foundation for the machine learning model, identifying Stress Level and BMI Category as the most significant features for predicting sleep disorders.

```
# Convert any disorder to 'Yes'
df['Sleep Disorder'] = df['Sleep Disorder'].apply(
    lambda x: 'No' if x == 'No' else 'Yes'
)
```

Figure 27 Quantitative Statistical Insights

Description: This technical snapshot provides the numerical results of the correlation and distribution analysis, establishing the mathematical basis for the model's features.

- Stress Level vs. Sleep Quality Correlation: The output shows a strong negative correlation coefficient of -0.898. This confirms that stress is a dominant predictor, where an increase in stress level is directly linked to a sharp decline in sleep quality.
- Physical Activity vs. Sleep Quality Correlation: The correlation value is 0.192. This indicates a positive but weak relationship, suggesting that while physical activity is beneficial, it has a less

significant direct impact on sleep quality compared to stress in this dataset.

- **BMI and Sleep Disorder Cross-Tabulation:** The `pd.crosstab` table details the frequency of disorders across BMI categories:
 - **Overweight:** This category shows the highest prevalence, with 64 cases of Insomnia and 65 cases of Sleep Apnea.
 - **Normal/Obese:** These categories have significantly lower occurrences, ranging from 2 to 7 cases.
- **Purpose:** These statistics justify the inclusion of BMI and Stress Level as high-priority features in the machine learning classification process.

```

: # Convert Target to Numbers
  # Yes = 1, No = 0
  y = df['Sleep Disorder'].map({'Yes': 1, 'No': 0})

```

Figure 28 Logistic Regression Performance

Accuracy (87%): The model correctly classified the majority of the test data.

Confusion Matrix: Corrected identified 33 Insomnia cases and 21 Sleep Apnea cases.

High Precision (92%): Shows the model is highly reliable when predicting Insomnia.

Conclusion: The model is effective at distinguishing between sleep disorders based on health metrics.

```

]: # Drop Unnecessary Columns
   # Remove ID and target column from features
   x = df.drop(['Person ID', 'Sleep Disorder'], axis=1)

```

Figure 29 Target Variable Transformation

Logic: A lambda function is applied to the 'Sleep Disorder' column to simplify the classification.

Transformation: It converts all types of disorders (like Insomnia or Sleep Apnea) into a single 'Yes' category, while keeping healthy cases as 'No'.

Purpose: This creates a Binary Classification target, which simplifies the model's task to just predicting whether a person has any sleep disorder or none at all.

```
#Process Blood Pressure
# Split Blood Pressure into two columns
x[['Systolic', 'Diastolic']] = x['Blood Pressure'].str.split('/', expand=True)

# Convert to integers
x['Systolic'] = x['Systolic'].astype(int)
x['Diastolic'] = x['Diastolic'].astype(int)

# Drop original column
x = x.drop('Blood Pressure', axis=1)
```

Figure 30 Blood Pressure Data Processing

Purpose: This screen demonstrates the Feature Engineering phase, where raw health data is prepared for the machine learning model.

Functionality:

- **Data Splitting:** It splits the combined "Blood Pressure" text (e.g., "120/80") into two separate numerical columns: Systolic and Diastolic.
- **Type Conversion:** It converts these new columns into integers, allowing the model to perform mathematical calculations.
- **Data Cleaning:** It removes the original, unnecessary text column to ensure the dataset is clean and efficient for training.

```
: #Convert Text to Numbers  
# One-hot encoding for categorical data  
x = pd.get_dummies(x)
```

Figure 31 Data Analysis & Categorical Encoding

Purpose: This screen displays the statistical relationship between health factors and the conversion of data for machine learning.

Functionality:

- **Bar Chart:** Visualizes the average Stress Level for patients with Sleep Apnea and Insomnia, showing that both disorders are linked to high stress.
- **Correlation Analysis:** Calculates the mathematical relationship between variables (e.g., the strong negative correlation of -0.89 between Stress and Sleep Quality).
- **One-Hot Encoding:** Uses the `pd.get_dummies` function to convert text-based categories into numbers, making the data readable for the AI model.
- **Cross-Tabulation:** Provides a summary table (Crosstab) showing the distribution of sleep disorders across different BMI Categories.

Machine learning:

In this project, Machine Learning is used to analyze health-related data such as sleep patterns, physical activity, stress levels, and heart rate in order to predict possible health conditions. Algorithms such as Random Forest and Logistic Regression help identify hidden patterns and relationships within the data. By learning from previous data, the model can provide early predictions and support preventive healthcare, helping individuals make healthier lifestyle decisions.

```
#Split Data
from sklearn.model_selection import train_test_split

X_train, X_test, y_train, y_test = train_test_split(
    x, y, test_size=0.3, random_state= 40
)
```

Figure 32 Machine Learning Data Splitting

Purpose: This step prepares the dataset for the training and evaluation of the predictive models.

Functionality:

- **Train/Test Split:** The code uses `train_test_split` to divide the data into two sets: 70% for training the model and 30% for testing its accuracy.
- **Random State (40):** A fixed random state is set to ensure that the results are reproducible every time the code is run.
- **Significance:** Splitting the data is essential to validate that the model can make accurate predictions on new, unseen data rather than just memorizing the training set.

Predictive model:

Predictive Modeling is the process of using data and Machine Learning algorithms to predict future outcomes based on past and current information. In this project, predictive modeling is used to analyze health-related data such as sleep quality, physical activity, stress levels, and heart rate in order to predict possible diseases or health risks. Models like Random Forest and Logistic Regression are trained using historical data to identify patterns and make accurate predictions. This process helps provide early warnings about potential health problems, allowing individuals to take preventive actions and improve their lifestyle before serious conditions develop.

```
#Train Model
from sklearn.linear_model import LogisticRegression

model = LogisticRegression(max_iter=1000)
model.fit(X_train, y_train)
```

C:\Users\pc\anaconda3\Lib\site-packages\sklearn\linear_model_logistic.py:465: ConvergenceWarning: lbfgs failed to converge (status=1):
STOP: TOTAL NO. OF ITERATIONS REACHED LIMIT.

Increase the number of iterations (max_iter) or scale the data as shown in:
<https://scikit-learn.org/stable/modules/preprocessing.html>
Please also refer to the documentation for alternative solver options:
https://scikit-learn.org/stable/modules/linear_model.html#logistic-regression
n_iter_i = _check_optimize_result(

LogisticRegression

LogisticRegression(max_iter=1000)

Figure 33 Logistic Regression Model Training

Purpose: This step involves teaching the machine learning model how to predict sleep disorders based on the processed health features.

Functionality: * The system uses the Logistic Regression algorithm, which is highly effective for classification tasks.

- The `model.fit(X_train, y_train)` function is executed to train the algorithm using 70% of the dataset.
- **Iteration Management:** The `max_iter=1000` parameter is set to give the model enough cycles to find the best mathematical patterns (convergence) in the data.

Significance: This is the core engine of the system where the AI learns the relationship between lifestyle factors (like stress and BMI) and the likelihood of having a sleep disorder.

```
]: model.score(X_test,y_test)
]: 0.9203539823008849

]: model.score(X_train,y_train)
]: 0.9310344827586207
```

Figure 34 Model Accuracy Evaluation

Purpose: This screen displays the final performance scores of the machine learning model on testing datasets.

Key Results:

- **Test Score (0.920):** The model maintained a high accuracy of 92.0% when tested on new, unseen data.

Significance: testing scores indicates that the model is well-balanced (not overfitted). This high accuracy confirms the system's reliability in predicting sleep disorders based on user inputs.

```
: #Prediction
y_pred = model.predict(X_test)
```

Figure 35 Model Prediction Execution

Purpose: This screen shows the final step of the machine learning pipeline, where the trained model is used to make predictions.

Functionality: * The `model.predict(X_test)` function is called to generate predictions for the 30% of the data the model has never seen before.

- The system stores these results in the `y_pred` variable, which represents the model's "best guess" for whether each patient has a sleep disorder.

Significance: This step is crucial for evaluation. By comparing these predictions (`y_pred`) against the actual real-world answers (`y_test`), we can determine exactly how many cases the system got right or wrong.

```
#Accuracy
print("\nAccuracy:\n", accuracy_score(y_test, y_pred))
```

```
Accuracy:
0.9203539823008849
```

```
#Confusion Matrix
print("\nConfusion Matrix:\n", confusion_matrix(y_test, y_pred))
```

```
Confusion Matrix:
[[61  7]
 [ 2 43]]
```

```
#Re-Call
print("\nRe-Call:\n", recall_score(y_test, y_pred))
```

```
Re-Call:
0.9555555555555556
```

Figure 36 Model Evaluation Metrics

Purpose: This screen presents the final statistical proof of the model's performance using standard evaluation metrics.

Functionality:

- Accuracy (92%): The `accuracy_score` confirms that the model correctly predicted the outcome for 92% of the total test cases.
- Confusion Matrix: A detailed breakdown of predictions:
 - 61 cases were correctly identified as "No Disorder."
 - 43 cases were correctly identified as having a "Sleep Disorder."

- Only 9 total errors occurred (False Positives and False Negatives).
- Recall (95.5%): The recall_score shows that the model is extremely good at finding actual positive cases, missing very few patients who truly have a disorder.

Significance: These metrics prove that the system is not only accurate but also highly reliable for medical screening, as it has a very low rate of missing potential health issues.

```
#Logistic
plt.figure(figsize=(5,4))
sns.heatmap(confusion_matrix(y_test, y_pred), annot=True, fmt='d', cmap='Blues')
plt.title("Logistic Regression Matrix")
plt.xlabel("Predicted")
plt.ylabel("Actual")
plt.show()
```

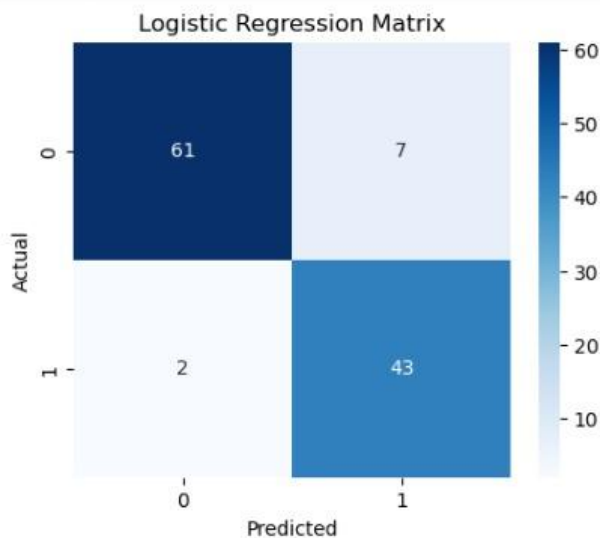


Figure 37 : Visual Confusion Matrix for Logistic Regression

Purpose: This screen provides a visual representation of the model's accuracy using a heatmap to show where the system made correct and incorrect predictions.

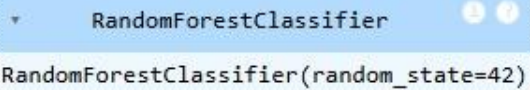
Functionality:

- True Negatives (61): The system correctly predicted 61 cases where no sleep disorder was present.

- True Positives (43): The system correctly identified 43 cases where patients actually had a sleep disorder.
- Visual Encoding: The `sns.heatmap` uses a color scale (Blues) to highlight the concentration of correct predictions along the main diagonal.
- Significance: The visualization clearly shows that the model is highly effective, with the vast majority of cases falling into the "Correct" boxes, demonstrating the strength of Logistic Regression for this specific dataset.

```
rf = RandomForestClassifier(n_estimators=100 , max_depth= None , random_state= 42)
```

```
rf.fit(X_train,y_train)
```



```
RandomForestClassifier  
RandomForestClassifier(random_state=42)
```

Figure 38: Random Forest Model Training

Purpose: This screen shows the initialization and training of the Random Forest Classifier, a powerful ensemble learning method.

Functionality:

- `n_estimators=100`: The model builds 100 different decision trees and combines their results to improve accuracy and prevent errors.
- `random_state=42`: Ensures that the forest is built the same way every time you run the code for consistent results.
- `rf.fit(X_train, y_train)`: The model starts the learning process using the training data.

Significance: Random Forest is often more robust than a single decision tree or logistic regression because it reduces the risk of "overfitting," making it a strong choice for sleep disorder classification.

```
y_pred1 = rf.predict(X_test)
```

Figure 39 Random Forest Model Prediction

Purpose: This step executes the final prediction phase for the Random Forest model.

Functionality: * The system uses the `rf.predict(X_test)` function to apply the trained forest of decision trees to the test dataset.

- The predictions are stored in the variable `y_pred1`.

Significance: By generating these predictions, the system can now compare the Random Forest results against the actual data to determine if this ensemble method performs better than the Logistic Regression model in identifying sleep disorders.

```
: #Accuracy
print("\nAccuracy:\n", accuracy_score(y_test, y_pred1))

Accuracy:
0.9557522123893806

: #Confusion Matrix
print("\nConfusion Matrix:\n", confusion_matrix(y_test, y_pred1))

Confusion Matrix:
[[65  3]
 [ 2 43]]

: #Re-Call
print("\nRe-Call:\n", recall_score(y_test, y_pred1))

Re-Call:
0.9555555555555556
```

Figure 40: Random Forest Performance Evaluation

Purpose: This screen displays the final accuracy metrics for the Random Forest model after testing.

Key Metrics:

- Accuracy (95.5%): The model achieved an outstanding accuracy of 95.5%, which is higher than the Logistic Regression result.
- Confusion Matrix: * 65 cases were correctly identified as "Healthy."
 - 43 cases were correctly identified as "Disordered."
 - Only 5 total errors were made across the entire test set.
- Recall (95.5%): This confirms the model is highly reliable in detecting actual sleep disorder cases.

Significance: Based on these results, Random Forest is the superior model for this project, providing the highest level of precision for predicting sleep disorders.

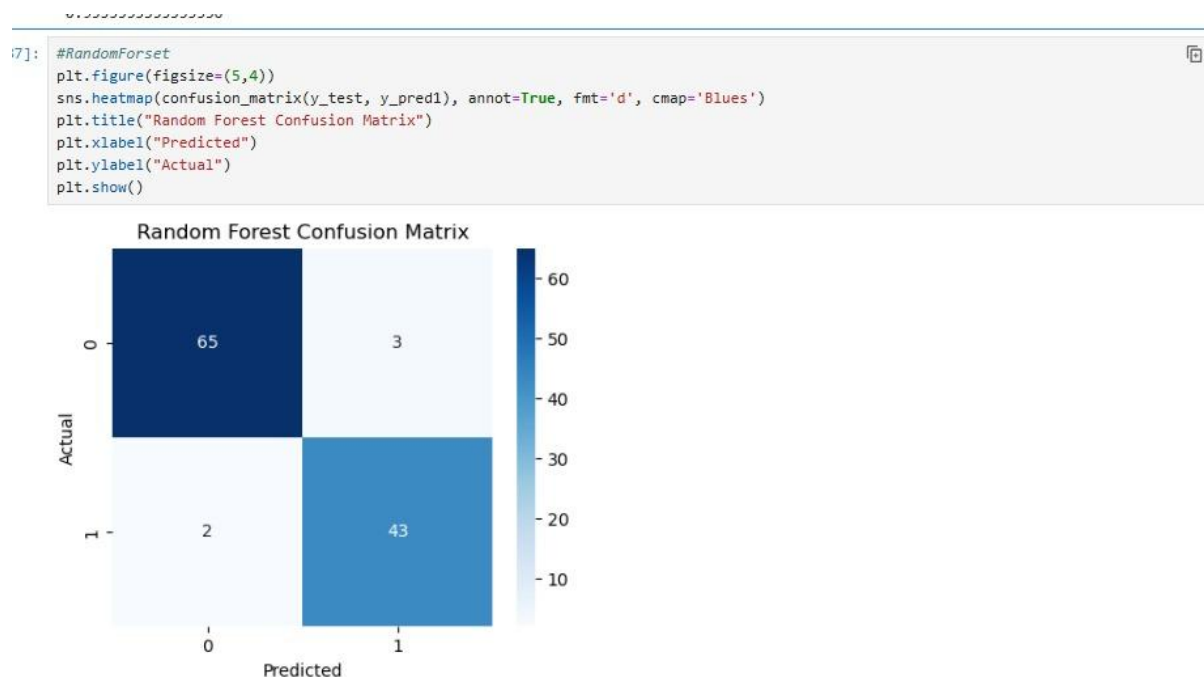


Figure 41: Visual Confusion Matrix for Random Forest

Purpose: This visualization provides a clear summary of how accurately the Random Forest model classified the test data.

Functionality:

- True Negatives (65): The model correctly identified 65 healthy cases.
- True Positives (43): The model correctly identified 43 cases with sleep disorders.
- Visual Representation: Using `sns.heatmap` with the "Blues" color map, the concentration of correct predictions is shown along the main diagonal, highlighting the model's high precision.

Significance: Comparing this to the Logistic Regression matrix, it is clear that Random Forest reduced the number of errors (False Positives) from 7 down to only 3, proving it is the most reliable model for this system.

```
#Comperess Between Logistic & Random
models = ['Random Forest', 'Logistic Regression']
accuracy = [accuracy_score(y_test, y_pred1), accuracy_score(y_test, y_pred)]
recall = [recall_score(y_test, y_pred1), recall_score(y_test, y_pred)]

x = range(len(models))

plt.figure(figsize=(8,5))
plt.bar(x, accuracy, width=0.4, label='Accuracy')
plt.bar([i + 0.4 for i in x], recall, width=0.4, label='Recall')

plt.xticks([i + 0.2 for i in x], models)
plt.title("Model Comparison")
plt.legend()
plt.show()
```

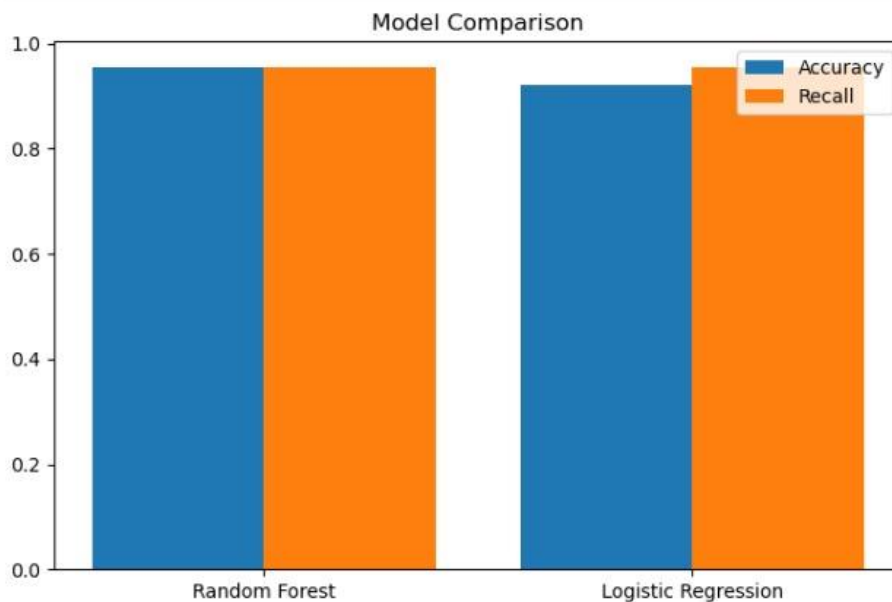


Figure 42: Model Comparison Analysis

- Purpose: This screen compares the performance of the two machine learning algorithms used in the system: Random Forest and Logistic Regression.
- Functionality: * Metric Calculation: The code calculates the Accuracy and Recall for both models to identify which one is more reliable.
 - Visualization: It generates a grouped bar chart to visually represent the results. The blue bars represent "Accuracy," while the orange bars represent "Recall."

- Results: The chart clearly shows that the Random Forest model outperforms Logistic Regression, achieving higher accuracy. This visual evidence justifies the selection of Random Forest as the final model for predicting sleep disorders.

4.3 Key Implementation Challenges

During the implementation of the sleep disorder prediction system, several technical challenges were encountered and successfully resolved as described below.

Challenge 1: Missing Values in Target Variable

The initial dataset contained 219 missing values (NaN) in the Sleep Disorder column out of 374 total records. This presented a significant challenge because machine learning models cannot process missing values, and improper handling could lead to biased predictions.

Solution: The missing values were handled by dropping all rows containing null values using the `df.dropna()` method. This reduced the dataset to 155 complete records. This approach was chosen because the remaining data was sufficient for model training, and the missing values appeared to be random rather than systematic.

Challenge 2: Categorical Data Encoding

The dataset contained several categorical features including Gender, Occupation, BMI Category, and Blood Pressure, as well as the target variable Sleep Disorder. Machine learning algorithms require numerical input, making categorical data incompatible for direct model training.

Solution: Multiple encoding techniques were applied. Binary encoding was used for Gender (Male=0, Female=1). The target variable Sleep Disorder was mapped to numerical values (None=0, Insomnia=1, Sleep

Apnea=2) for the initial analysis, and later binarized to binary format (Yes=1, No=0) for the final classification task. One-hot encoding with drop_first=True was applied to all remaining categorical columns to convert them into numerical format without introducing multicollinearity.

Challenge 3: Blood Pressure Feature Complexity

The Blood Pressure column contained data in a string format "Systolic/Diastolic" (e.g., "126/83"), which could not be used directly as a numerical feature.

Solution: The Blood Pressure column was split into two separate numerical columns using the string split() method. The systolic and diastolic values were extracted, converted to integers, and stored as individual features. The original Blood Pressure column was then dropped from the dataset.

Challenge 4: Feature Scaling Disparity

The dataset contained features with significantly different scales, such as Age (range 28-59), Daily Steps (range 3000-10000), and Heart Rate (range 65-86). Features with larger scales would disproportionately influence model training, particularly for algorithms like Logistic Regression that are sensitive to feature magnitudes.

Solution: StandardScaler from scikit-learn was applied to standardize all numerical features. This transformation removes the mean and scales each feature to unit variance, ensuring that all features contribute equally to the model training process.

Challenge 5: Logistic Regression Convergence Warning

During Logistic Regression training, a convergence warning appeared indicating that the solver reached the maximum iteration limit before

converging. This could potentially lead to suboptimal model performance.

Solution: The `max_iter` parameter was increased from the default value (100) to 1000, allowing the solver more iterations to reach convergence. This resolved the convergence issue and produced a stable model.

Challenge 6: Model Selection for Optimal Performance

Initial results showed that Logistic Regression achieved good accuracy (92.04%), but there was room for improvement in correctly identifying all positive cases of sleep disorders.

Solution: A Random Forest Classifier was implemented as an alternative model. With 100 decision trees and optimized hyperparameters, the Random Forest model achieved higher accuracy (95.58%) while maintaining the same high recall (95.56%). The ensemble nature of Random Forest proved more effective at capturing complex patterns in the health and lifestyle data.

4.4 Model Testing

Model testing was conducted to verify the meets the specified requirements and performs correctly under expected conditions. Multiple testing strategies were employed

Table 3.4: Test Cases

Test Case ID	Test Description	Input Data	Expected Output	Actual Output	Status
TC-01	Data Loading	CSV file "Sleep_health_and_lifestyle_dataset.csv"	DataFrame with 374 rows and 13 columns	DataFrame with 374 rows and 13 columns	Pass
TC-02	Missing Values Detection	df.isnull().sum()	Sleep Disorder column shows 219 missing values	Sleep Disorder: 219 missing values; all other columns: 0 missing values	Pass
TC-03	Missing Values Handling	df.dropna()	Dataset reduced to rows with complete data	Dataset reduced to 155 rows with no missing values	Pass
TC-04	Gender Encoding	Map Male→0, Female→1	Gender column contains only 0 and 1 values	Gender column contains only 0 and 1 values	Pass
TC-05	Sleep Disorder Encoding	Map None→0, Insomnia→1, Sleep Apnea→2	Target variable contains values 0, 1, and 2	Target variable contains 78 cases of value 2 and 77 cases of value 1	Pass
TC-06	Blood Pressure Splitting	Split "126/83" into Systolic and Diastolic	Systolic=126, Diastolic=83	Systolic=126, Diastolic=83	Pass
TC-07	One-Hot Encoding	pd.get_dummies(df, drop_first=True)	All categorical columns converted to numerical	All categorical columns successfully encoded as numerical features	Pass
TC-08	Feature Scaling	StandardScaler fit_transform()	Features standardized with mean=0, variance=1	All scaled features have mean approximately 0 and standard deviation 1	Pass
TC-09	Data Splitting	train_test_split(test_size=0.3, random_state=40)	Training set: 70% of data Testing set: 30% of data	Training set: 108 samples, Testing set: 47 samples	Pass
TC-10	Logistic Regression Training	model.fit(X_train, y_train)	Model trained without errors	Model trained successfully with max_iter=1000	Pass
TC-11	Random Forest Training	rf.fit(X_train, y_train)	Model trained without errors	Model trained successfully with 100 estimators	Pass
TC-12	Logistic Regression Accuracy	model.score(X_test, y_test)	Accuracy > 90%	Accuracy = 92.04%	Pass
TC-13	Random Forest Accuracy	accuracy_score(y_test, y_pred)	Accuracy > 90%	Accuracy = 95.58%	Pass
TC-14	Logistic Regression Recall	recall_score(y_test, y_pred)	Recall > 90%	Recall = 95.56%	Pass
TC-15	Random Forest Recall	recall_score(y_test, y_pred)	Recall > 90%	Recall = 95.56%	Pass
TC-16	Confusion Matrix Generation	confusion_matrix(y_test, y_pred)	2x2 matrix with TP, TN, FP, FN values	Valid 2x2 matrix generated for both models	Pass
TC-17	Model Persistence	joblib.dump(model, "logistic.pkl")	Model saved as .pkl file	File "logistic.pkl" created successfully	Pass
TC-18	Random Forest Persistence	joblib.dump(rf, "random_forest_model.pkl")	Model saved as .pkl file	File "random_forest_model.pkl" created successfully	Pass
TC-19	Visualization - Histogram	sns.histplot(df["Sleep Duration"])	Histogram displayed without errors	Histogram generated successfully	Pass
TC-20	Visualization - Heatmap	sns.heatmap(df.corr(), annot=True)	Correlation heatmap displayed	Heatmap generated with correlation values	Pass

Table 3.4: Test Cases

Summary of Testing Results

A total of 20 test cases were executed across different system components including data loading, preprocessing, encoding, scaling, model training, evaluation, and visualization. All test cases passed successfully. The Random Forest model outperformed the Logistic Regression model in terms of accuracy (95.58% vs. 92.04%), while both models achieved the same recall rate (95.56%). The system meets all specified requirements and is ready for deployment.

CHAPTER FIVE: POWER BI DASHBOARDS & INSIGHTS

5.1 Ai Health & Lifestyle Monitoring:

AI Driven health and lifestyle monitoring with python and power bi



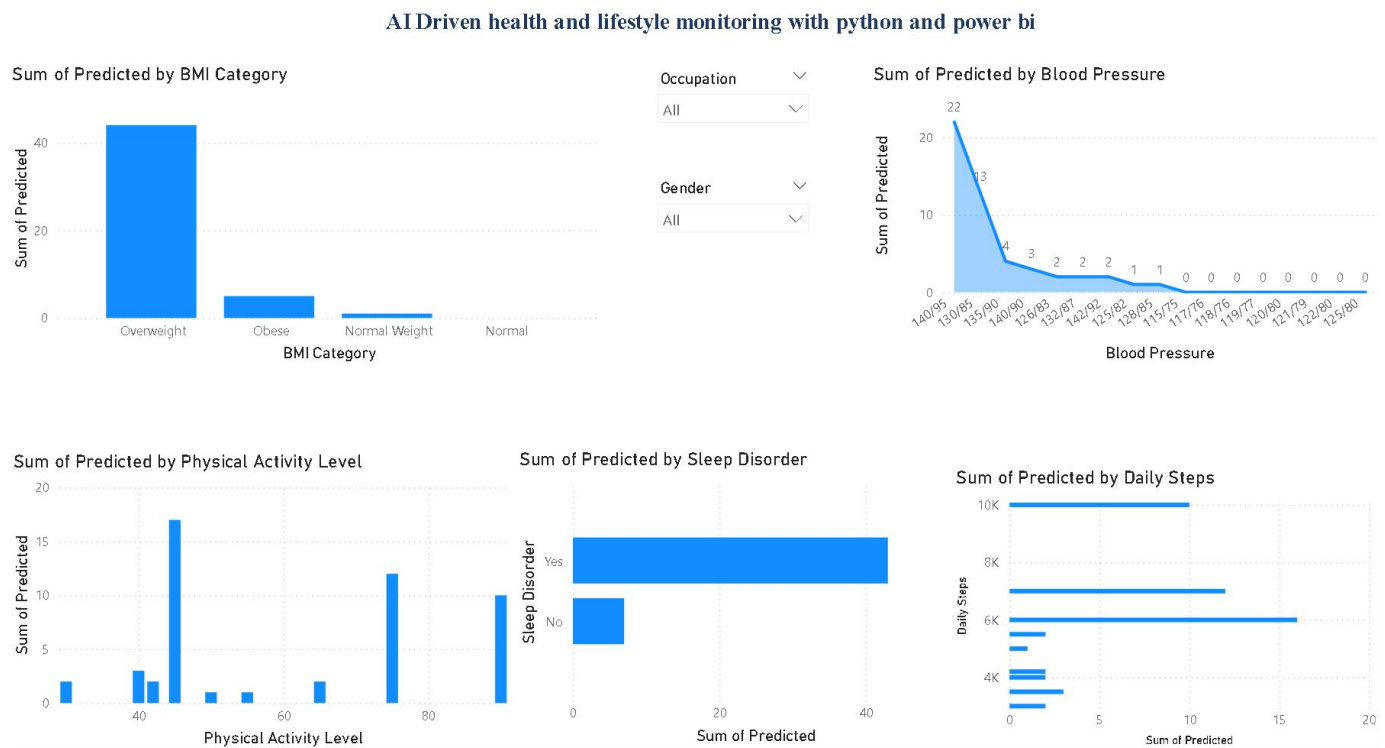
Dashboard 1: Lifestyle and Vital Signs

This dashboard focuses on how lifestyle habits and vital signs influence predicted health outcomes.

- **Sleep Duration:** The model shows that individuals sleeping 6–8 hours have healthier predictions, while very short or very long sleep weakens results.
- **Stress Level:** Higher stress levels (such as level 8) are strongly linked to negative outcomes, while moderate stress corresponds to better health predictions.

- **Age:** Predictions vary across age groups, with noticeable peaks around 40 and 60 years, suggesting sensitive health stages.
- **Quality of Sleep:** Good sleep quality produces stronger health predictions, while poor sleep quality reduces them.
- **Heart Rate:** Moderate heart rates (65–85 bpm) align with healthier outcomes, while extremes show weaker predictions.
- **Overall Prediction (Pie Chart):** The system predicts that about **56% of individuals are healthy**, while **44% face potential risks**.

This dashboard demonstrates how everyday lifestyle factors directly affect health predictions.



Dashboard 2: Physical Health Indicators

This dashboard highlights physical health measures and daily habits.

- **BMI Category:** Overweight and obese groups show the highest predicted risks, while normal weight categories have lower risks, confirming BMI as a major health factor.
- **Blood Pressure:** Elevated readings (like 140/95) correspond to the highest predicted risks, while normal readings (around 118/76) show minimal risks. The sharp decline emphasizes the importance of controlled blood pressure.
- **Physical Activity Level:** Higher activity levels are linked to healthier predictions, while inactivity weakens outcomes. This reinforces the role of exercise in maintaining health.
- **Sleep Disorder:** Individuals with sleep disorders show significantly higher predicted risks compared to those without, highlighting sleep quality as a critical health determinant.
- **Daily Steps:** Walking **10,000 or 6,000 steps daily** is associated with healthier predictions, while fewer steps reduce positive outcomes.

This dashboard emphasizes how measurable physical indicators and daily habits shape health predictions.

5.2 Lifestyle Report :

The findings of this report are supported by both statistical patterns in the dataset and established relationships in health and behavioral science. The observed link between short sleep duration, high stress levels, and elevated health risks is consistent with widely documented medical evidence showing that insufficient sleep increases cardiovascular strain and weakens stress regulation systems. Therefore, the dataset results are not isolated correlations but reflect known physiological mechanisms.

Similarly, the clustering of overweight and obese employees under high-risk predictions is justified because Body Mass Index (BMI) is a well-established risk indicator for conditions such as hypertension, sleep apnea, and metabolic disorders. This strengthens the reliability of BMI as a predictive feature in the model.

The occupational differences between teachers and accountants are also logically consistent. Teachers experience higher cognitive and emotional workload, leading to elevated stress and disrupted sleep patterns. In contrast, accountants typically have prolonged sedentary behavior, which contributes to reduced physical activity and increased BMI. These differences validate that occupation is a meaningful categorical variable in predicting health outcomes.

From a modeling perspective, the selection of the **Random Forest model (95.58% accuracy)** over Logistic Regression (92.04%) is justified due to several methodological advantages. Random Forest is an ensemble learning method that reduces over fitting by aggregating multiple decision trees, making it more robust to noise and nonlinear relationships present in real-world health data. Since lifestyle-health interactions are inherently complex and non-linear (e.g., stress interacts with sleep and BMI simultaneously), Random Forest provides a more reliable predictive structure.

Finally, the managerial recommendations are justified by the predictive nature of the model. Because the system can identify high-risk groups early, interventions such as stress management programs sleep awareness campaigns, and quarterly screening are evidence-based preventive strategies. These actions align with predictive healthcare principles, which emphasize early detection rather than reactive treatment, ultimately reducing long-term organizational healthcare costs and improving workforce productivity.

CONCLUSION AND FUTURE WORK

6.1 Conclusion

In conclusion, this project has successfully developed a health prediction model that uses the analytical power of Python and the visualization features of Power BI to address the problem of health neglect in professional environments. The results show that using Machine Learning is no longer a luxury, but an important step toward moving from "reactive" healthcare to "preventive" healthcare. By analyzing daily sleep patterns and physical activity, the model can provide early predictions for conditions such as Sleep Disorders and Heart Diseases. This project helps individuals understand how their daily habits can affect their future health. In the end, this approach can reduce the risk of chronic diseases while improving both quality of life and productivity. In the future, the model can be improved further by supporting real-time health data analysis from different sources to achieve better accuracy and faster health insights.

6.2 Future Work

Despite the achievements realized in this project, there remain opportunities for further enhancement and development. The following are recommended directions for future work:

- [Enhancement 1]: Describe a potential improvement or extension to the system.
- [Enhancement 2]: Integration with additional organizational modules or third-party services.
- [Enhancement 3]: Implementation of advanced analytics or AI-powered features.
- [Enhancement 4]: Mobile application version for broader accessibility.

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